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V810i S2EX System Overview and Introduction

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Automated Board Inspection

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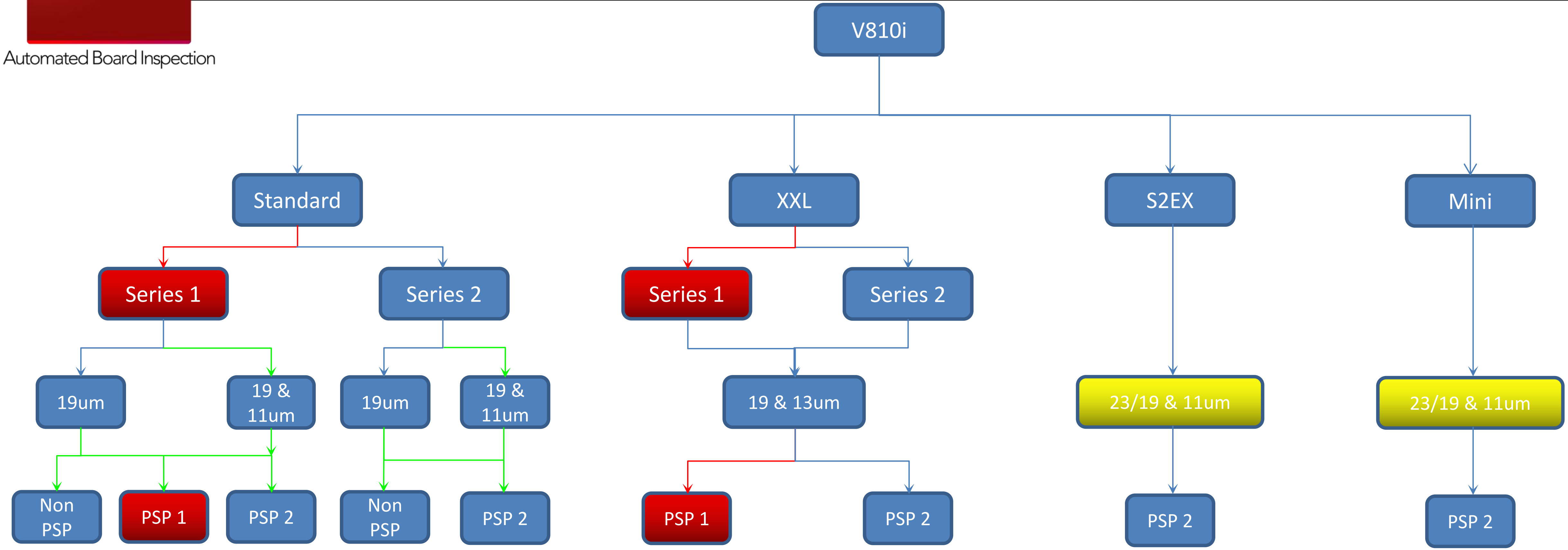
V810i S2EX System Overview

ViTrox V810i S2EX Automated X-ray Inspection System inspects the solder joints on printed circuit board assemblies using x-ray and digital reconstruction.

It identifies inadequate solder joints and misplaced or missing components.

Automated Board Inspection

V810i System Type Table Chart



Remark:

- Red = Obsolete
- Green = Optional
- Yellow = Programming option, 23+11um or 19+11um
- Blue = Basic package
- Std and XXL assembled with pnematic VM position change VM and S2EX and Mini running with motorized.

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V810i S2EX System Specifications

Board Dimensions	V810i S2EX
Maximum panel size	482mm X 610mm (19" x 24")
Minimum panel size	76mm X 76mm (3" x 3")
Maximum panel inspectable area	474mm X 610mm (18.7" x 24")
Maximum panel thickness	7 mm (276 mils)
Minimum panel thickness	0.5mm (20 mils)
Panel warp	Downside < 3.0mm; Upside < 1.5mm (PSP)
Maximum panel weight	4.5kg
Minimum panel weight	0.03kg
Clearance	
Board top clearance	50mm @ 23um resolution 38mm @ 19um resolution 11mm @ 11um resolution (calculated from **board** top surface)
Board bottom clearance	70mm
Panel edge clearance	3mm
Panel width tolerance	+/- 0.7mm
System resolution	11um/19um/23um
100% Press-fit testability	Yes (With PSP2 feature)
Maximum acceptable panel temperatures	40 Deg C
System Dimensions	
Footprint	1566mm (W) x 2145mm (D) x 1972mm (H)
Weight	~3500kg

Specifications

Systems	
System controller	Integrated controller with 8 Core Intel Xeon processors
Operating system	Windows 8 (64 bits)

Test Development Environment	
User interface	Microsoft Windows based software solution with easy-to-use GUI and password-protected user levels
Off-line test development software	Optional for off-line PC
Test sight developer	Optional software available to translate CAD data to ViTrox's format
Typical test development time	4 hours to 1.5 days to convert raw CAD file and develop application

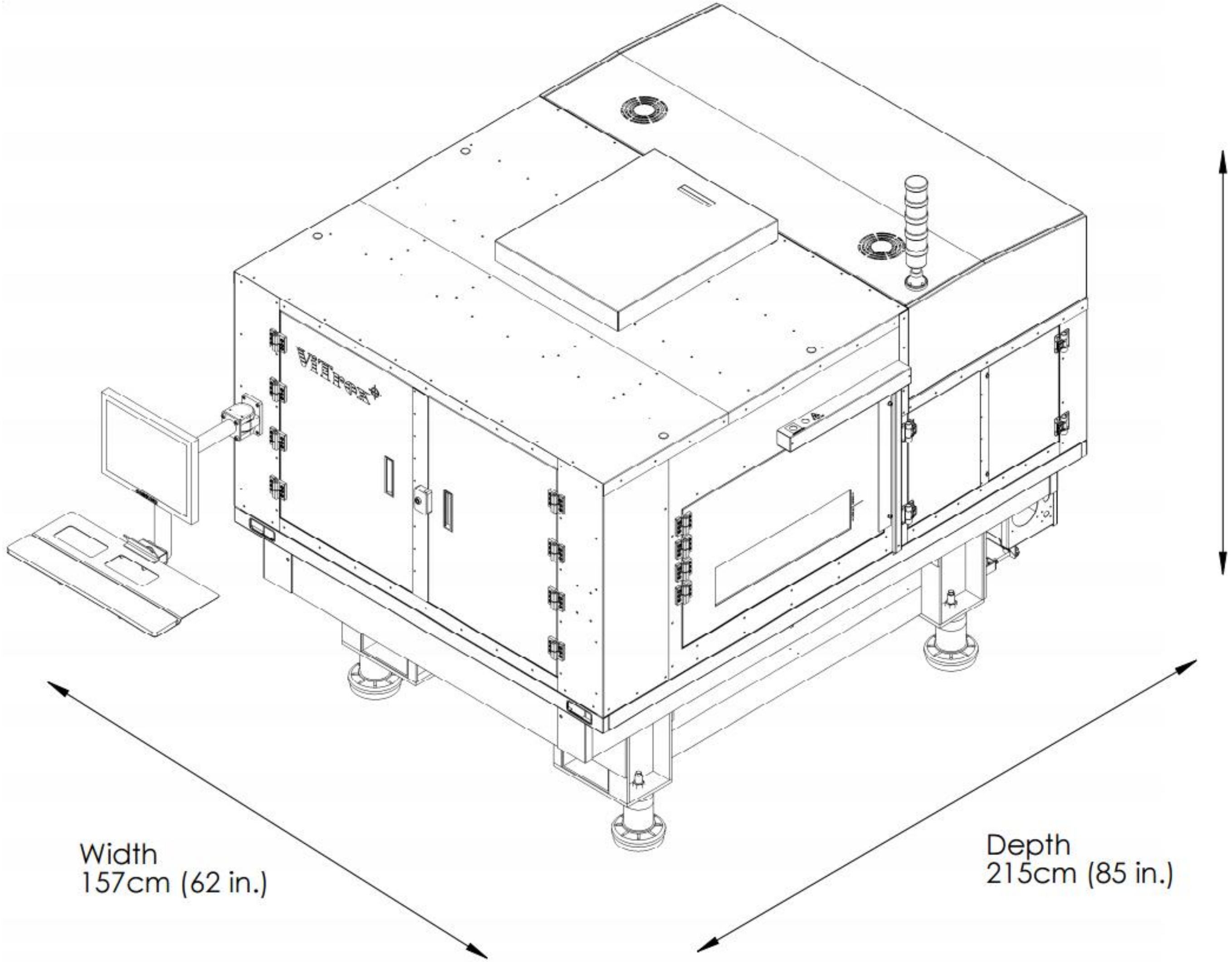
Line Integration	
Transport heights	865mm-1025mm
Line communication standard	SMEMA
Barcode readers	Compatible with most industry standard barcode readers

Performance Parameters *	
Total panel test cycle time	51.68cm^2/sec (8 in^2/sec) at 19um
Typical Image Acquisition Rate	500 - 1000 ppm
False call rates	
Minimum features detection capability	
Joint pitch[1]	0.3 mm and up
Short width[2]	0.045 mm
Solder thickness	0.0127 mm

Allowable panel characteristics **	
Maximum Panel size	482mm X 610mm (19"x24")
Minimum Panel size	76mm X 76mm (3" x 3")
Maximum panel inspectable area	474mm X 610mm (18.7" X 24")
Maximum Panel thickness	7mm(276mils)
Minimum Panel thickness	0.5mm (20mils)
Panel warp	Downside<3mm,upside< 1.5mm (PSP)
Maximum Panel weight	4.5kg
Minimum Panel weight	0.03kg
Board top clearance	50mm @ 23um resolution, 38mm @ 19um resolution 11mm @ 11um resolution ** Calculated from **board** top surface*
Board bottom clearance	70mm
Panel edge clearance	3.0mm
Panel width tolerance	+/-0.7mm
System resolution	11um, 19um, 23um
100% Press-fit testability	Yes (With PSP2 feature)
Maximum acceptable panel temperatures	40 Deg C

Power and environmental	
Voltage requirement	200 - 240 VAC three phase; 380 - 415 VAC three phase wye (± 5) (50Hz or 60 Hz)
Air requirement	552 kPA (80 psi) compressed air
System footprint (Width X Depth X Height)	1566mm X 2145mm X 1972mm
Total system weight	~3500kg

V810i S2EX System Dimension



V810i S2EX System Dimension

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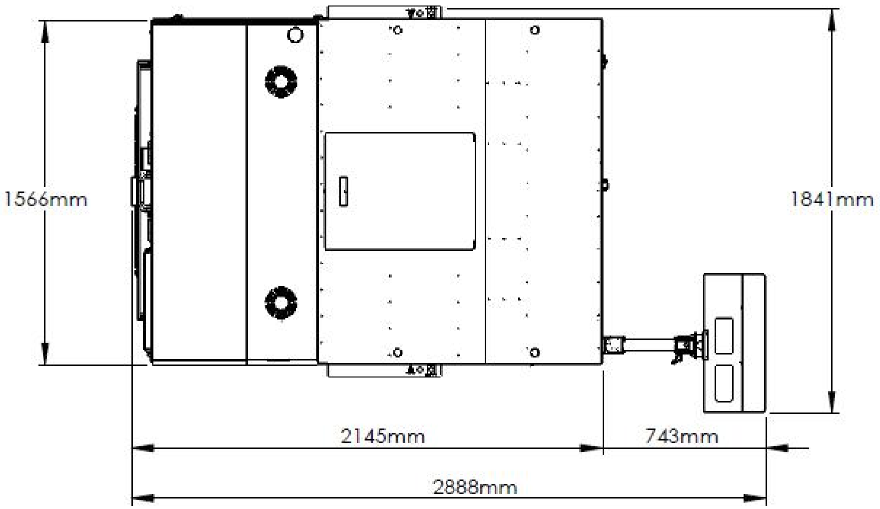

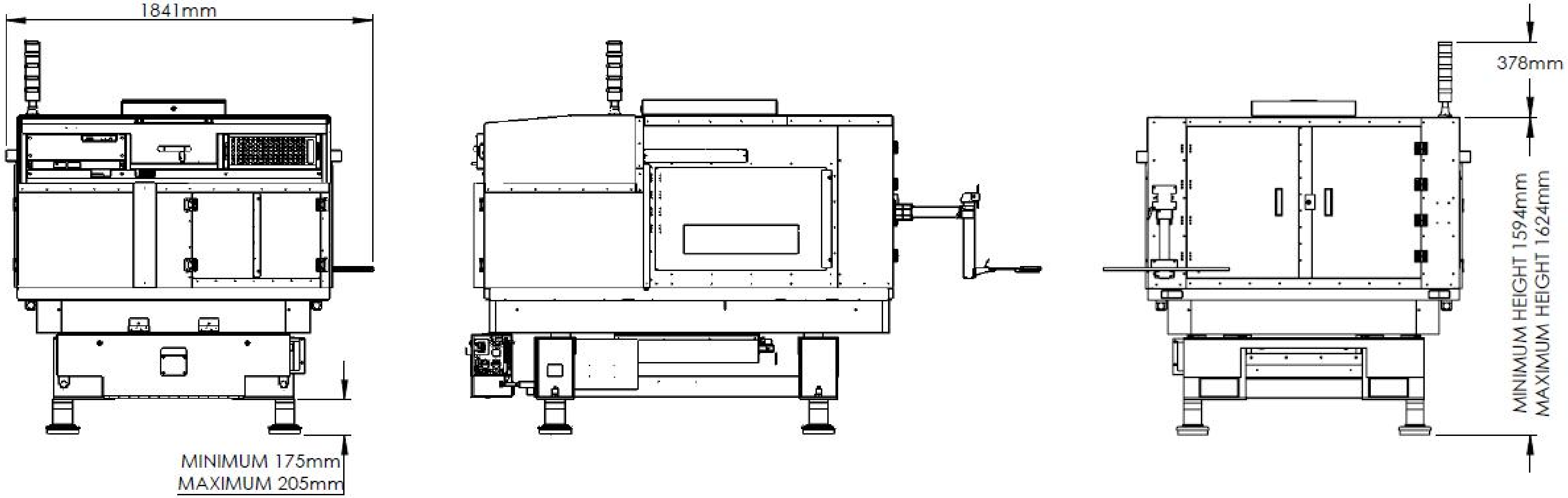

System Hardware Overview

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Front of System

1. Cabinet
2. Monitor Arm

   # Can be configured on either

   left or right side of cabinet
3. Operator Control Panel
4. Light Tower
5. Fork Truck Holes
6. Front door

System Hardware Overview

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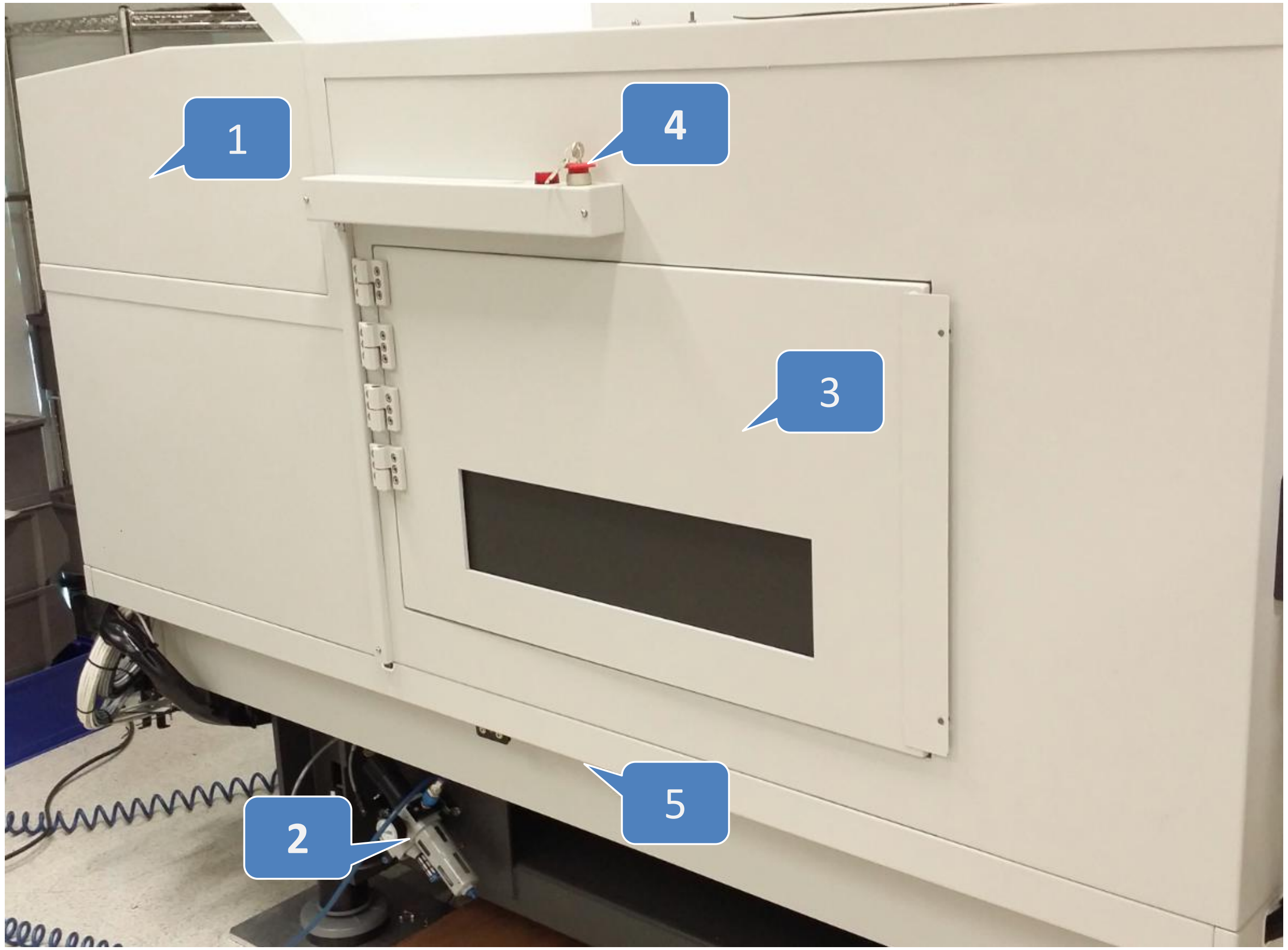
Left Side of System

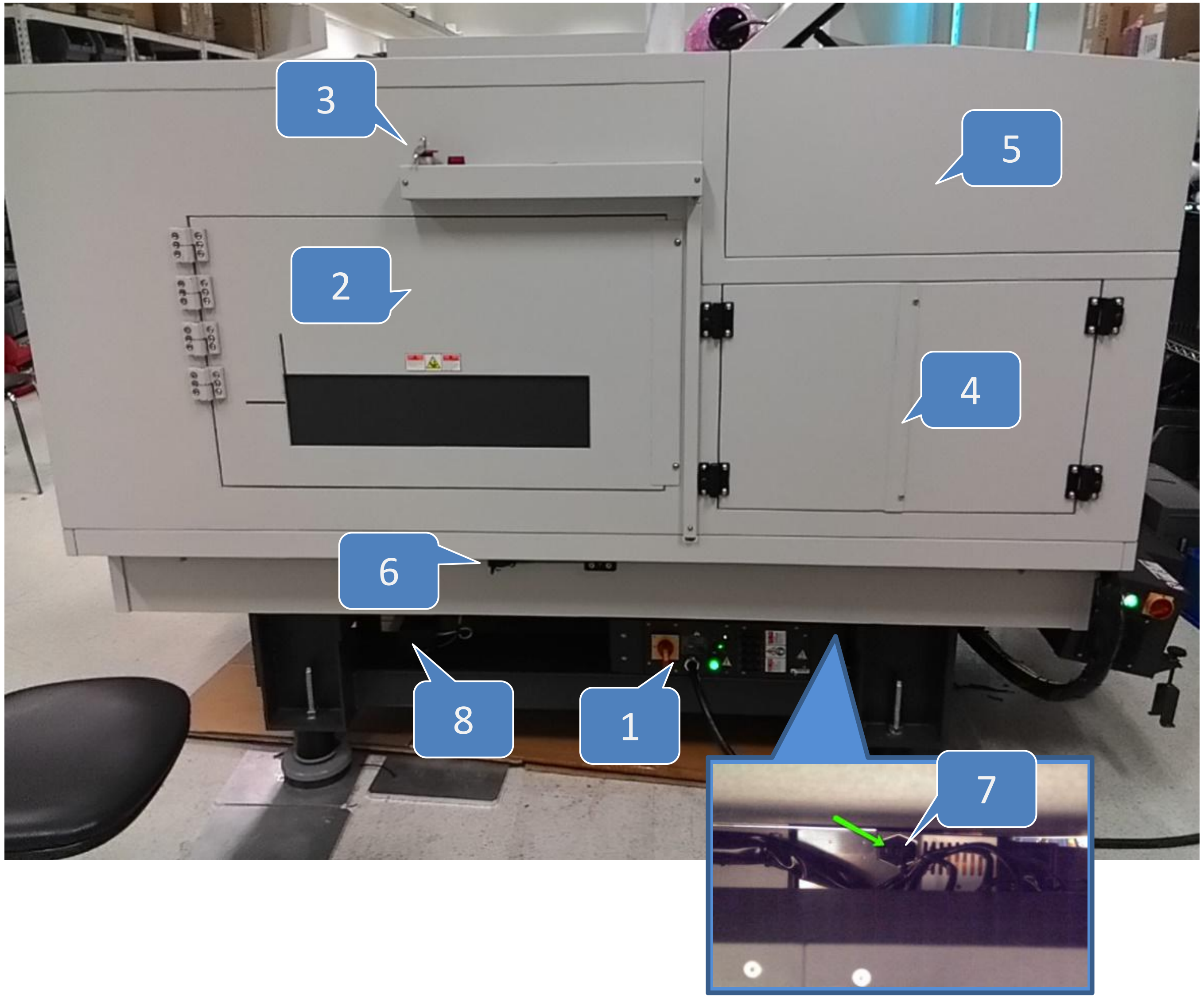
1. Controller Rack
2. Air pressure regulator
3. Left Outer Barrier
4. Emergency Off Switch – left side
5. External Smema Connection
   (Upstream Conveyor)



System Hardware Overview

Right Side of System

1. Power Distribution Unit (PDU)
2. Right Outer Barrier
3. Emergency Off Switch – Right side
4. Right Side Door
5. Server Controller Rack
6. External Smema Connection (Downstream)
7. Camera Trigger Board On/Off Switch
8. Break Out Board (BOB 2)

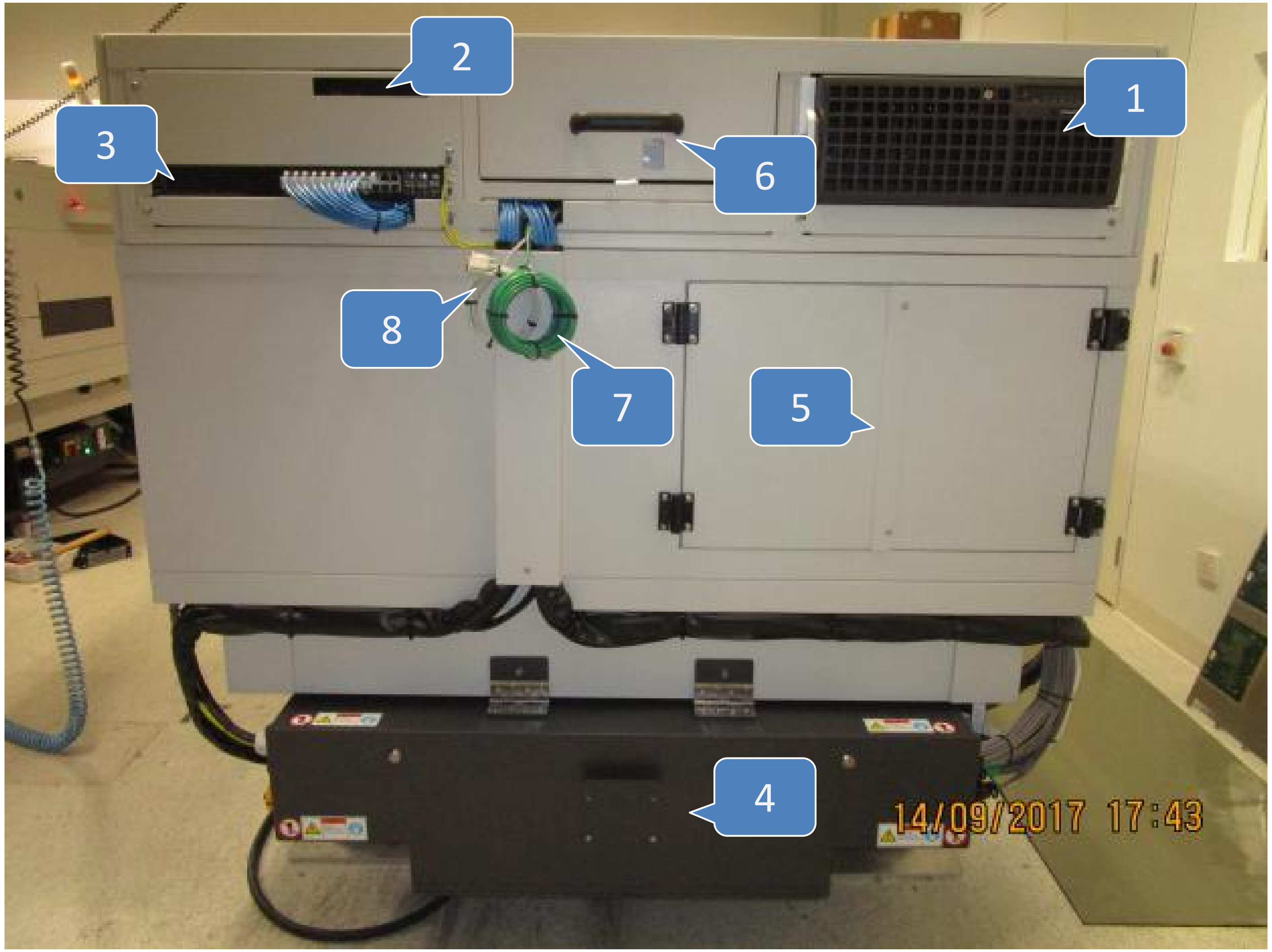


System Hardware Overview

Automated Board Inspection

Rear Side of System

1. S2 Super Micro Server
2. Atom PC
3. Gigabyte switch
4. System Status & Control Assembly (SSCA)
5. Rear Door
6. Server rack top cover handle and lock
7. Internet/Intranet LAN cable
8. Bar-code COM cable (RS232)



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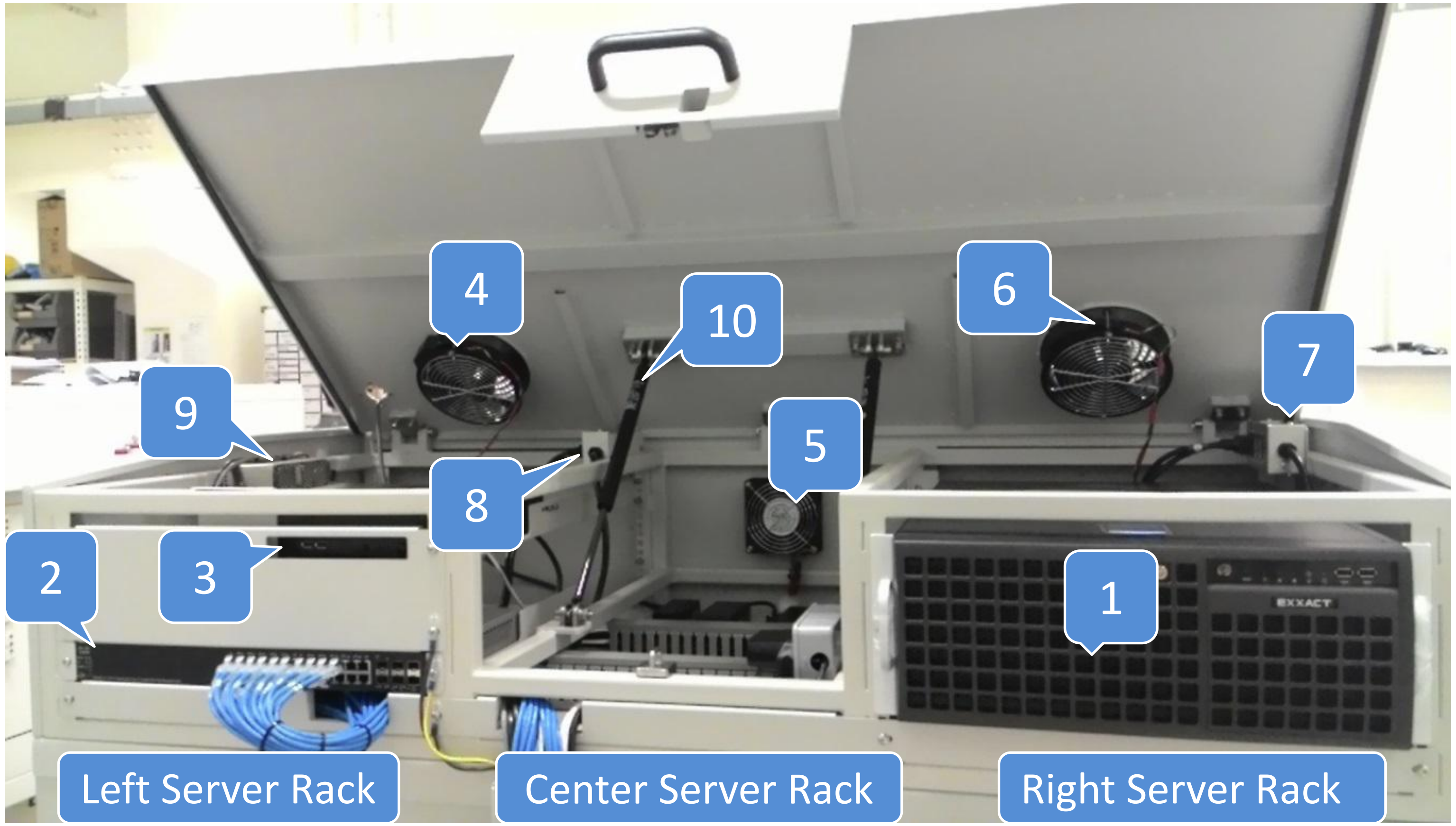
System Hardware Overview

Server Rack Inner View

1. S2 Super Micro Server
2. Cisco Gigabyte switch
3. Atom PC (Remote Motion Server)
4. Left IRP Fan
5. Center IRP Fan
6. Right IRP Fan
7. Left Power outlet box
8. Right Power outlet box
9. H-tube 24vdc Power Supply
10. Gas Dampers



System Hardware Overview

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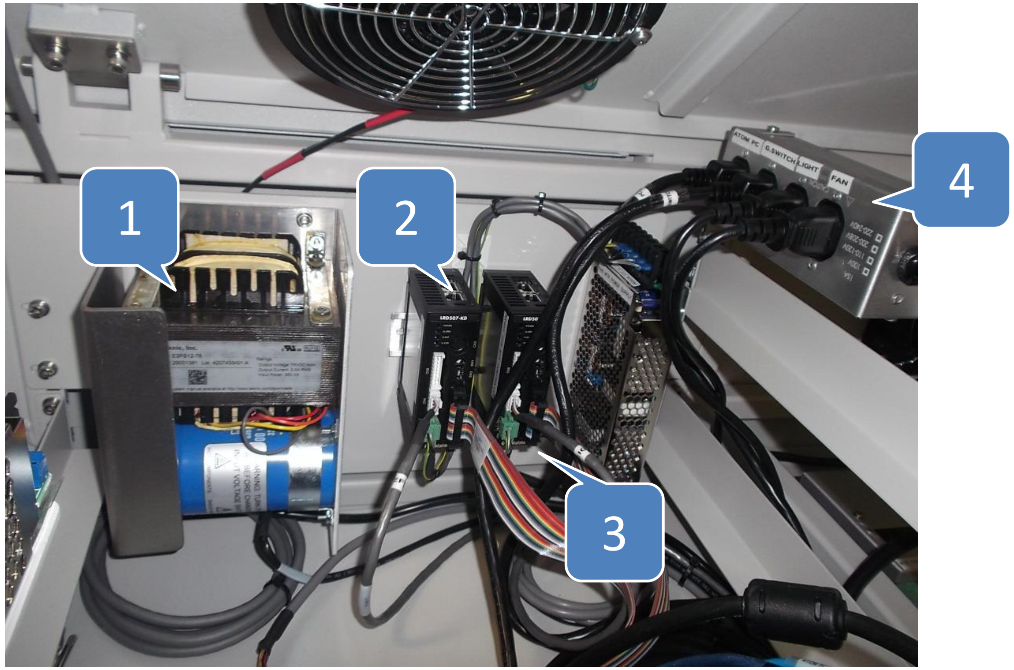
Server Rack Inner View

1. Z Axis Stepper Motor Power Supply
2. Left Filter Height Sensor Module Controller
3. Right Filter Height Sensor Module Controller
4. Filter Height Module Controller Power Supply
5. USB 2.0 A to Micro-B Cable 3M
   (Connected to Z Axis motor)



System Hardware Overview

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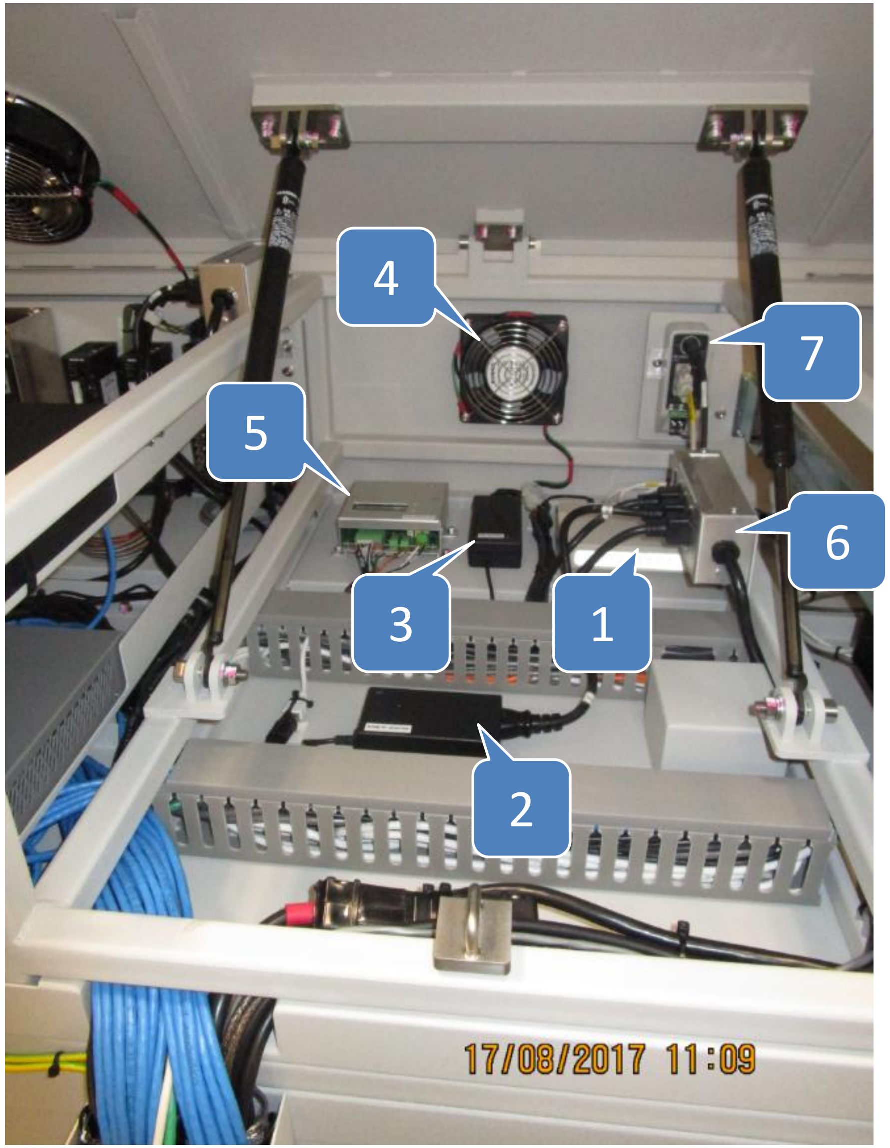
Center Server Rack Inner View (with PSP2)

1. POE Switch (with PSP Option only)
2. Atom PC Power Adaptor
3. POE Switch Power Adaptor
4. Center IRP Fan
5. PIP Slow Speed Controller
6. Center Server Rack Outlet Box
7. Keyance Bar-Code controller (optional)



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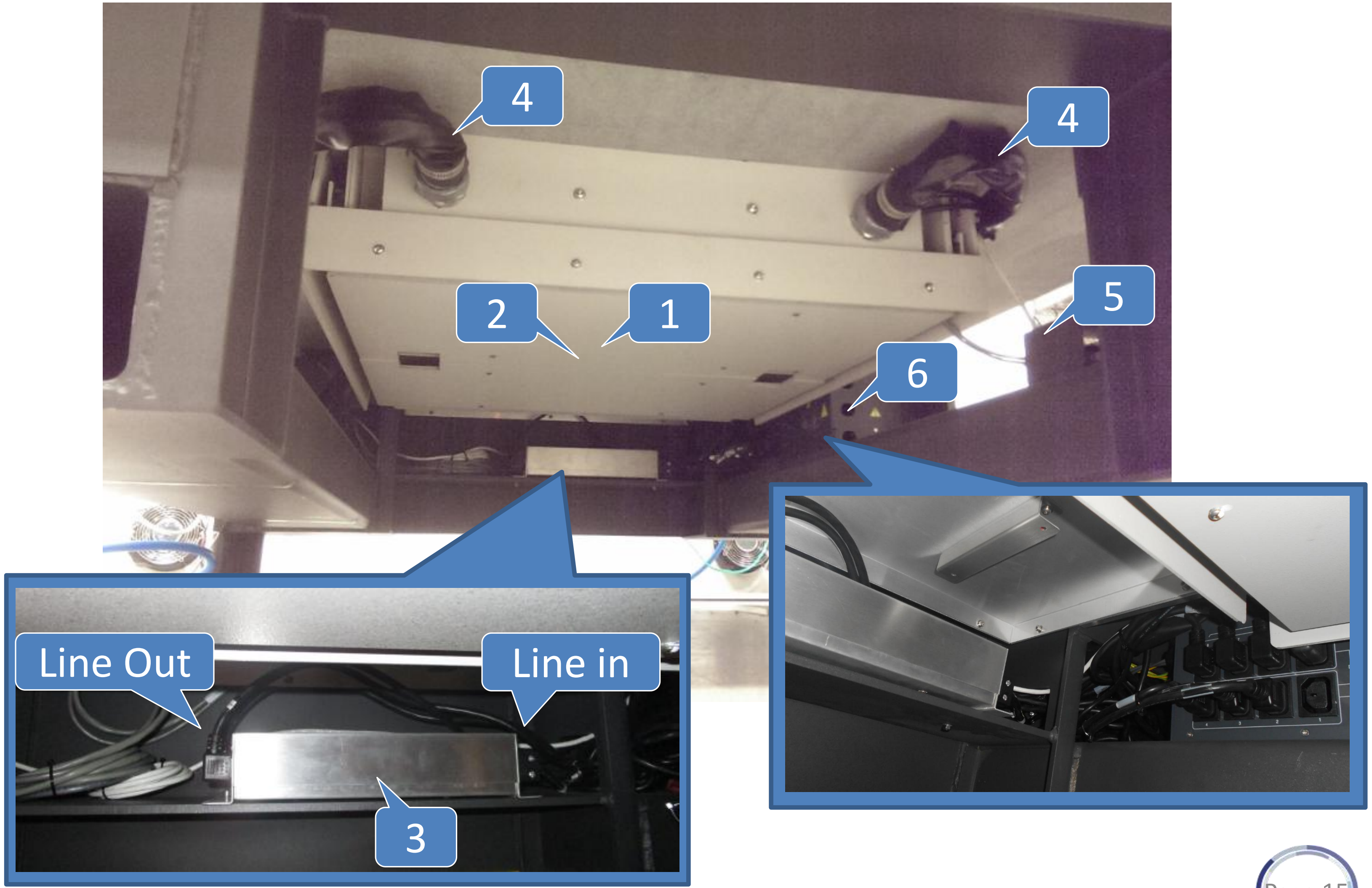
System Hardware Overview

Bottom View of System

1. Camera Array Enclosure with thumb lock
2. Camera Controller Assembly (Trigger board with power Supply)
3. 3 in 3 out Line Filter Box
4. Fire wires + Camera LAN cable
   - Fire wires route from trigger board
   - Camera LAN cables route from Gigabit Switch
5. Break Out Board (BOB 2)
6. Power Distribution Unit (PDU)



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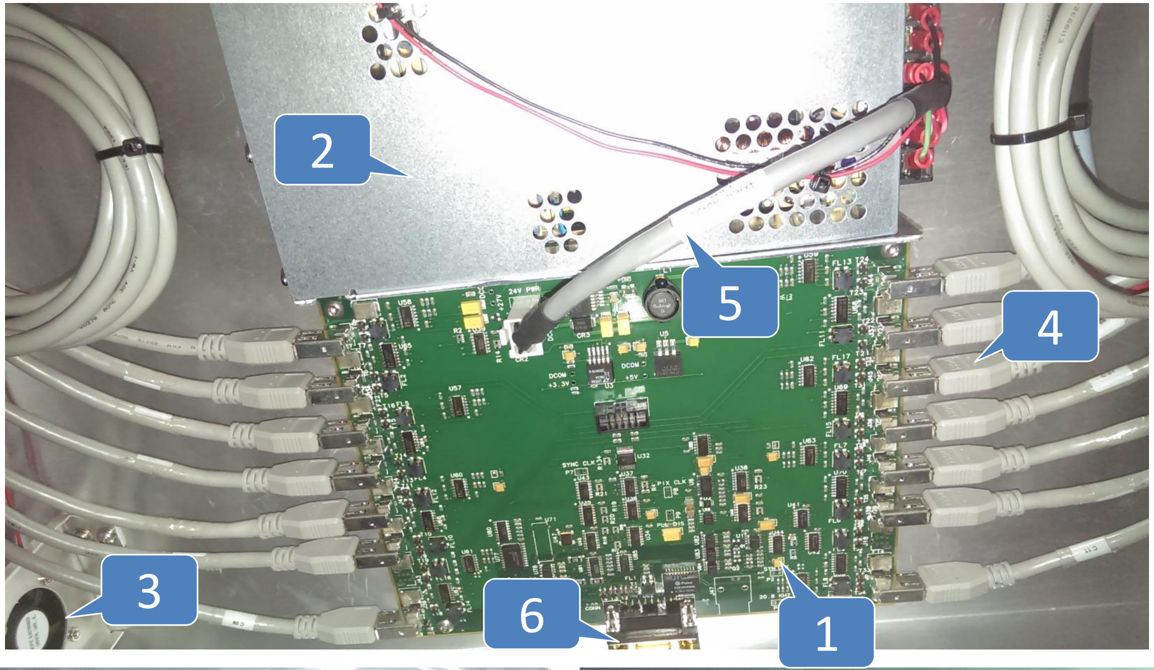
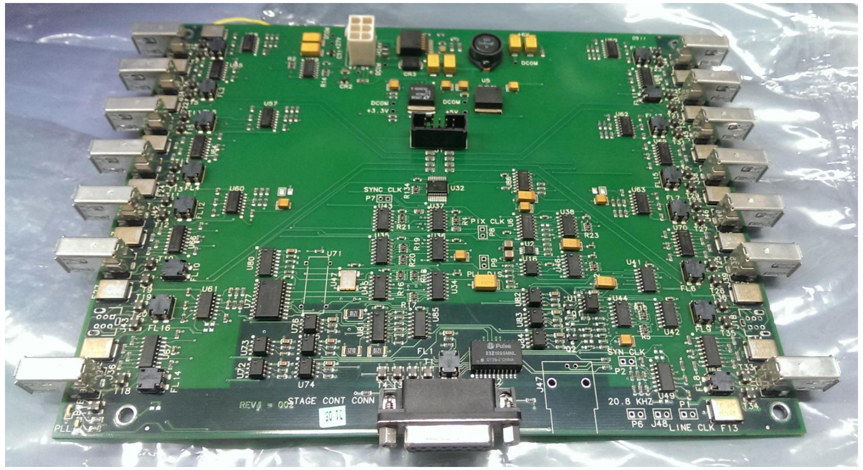
System Hardware Overview

Inner View of Camera Controller Assembly

1. Trigger Board
2. 24vDC LAMBA power supply
3. Cooling fan
4. Fire wires x 14 (No orientation needed)
5. Power cable
6. Trigger cable





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System Hardware Overview

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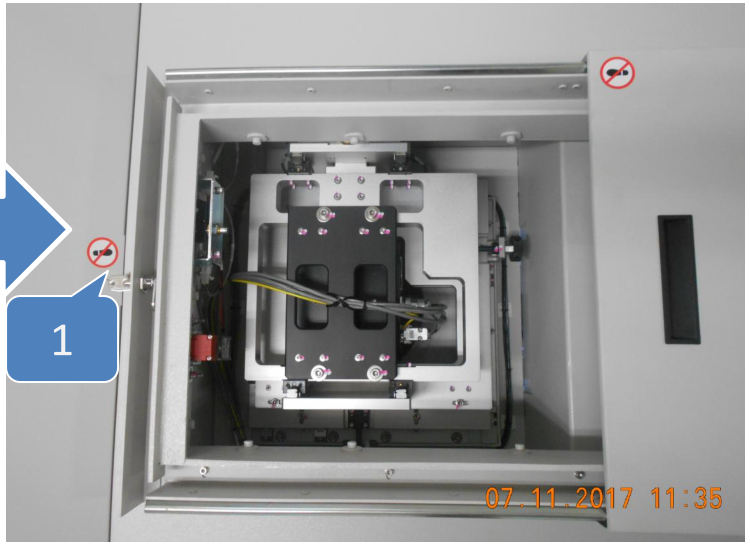
Top Access Service Door of System

1. Top Access Door (Tube Enclosure) – Same key as SSCA lock.

2. X-Ray Tube

3. X-Ray tube with mounting bracket



System Hardware Overview

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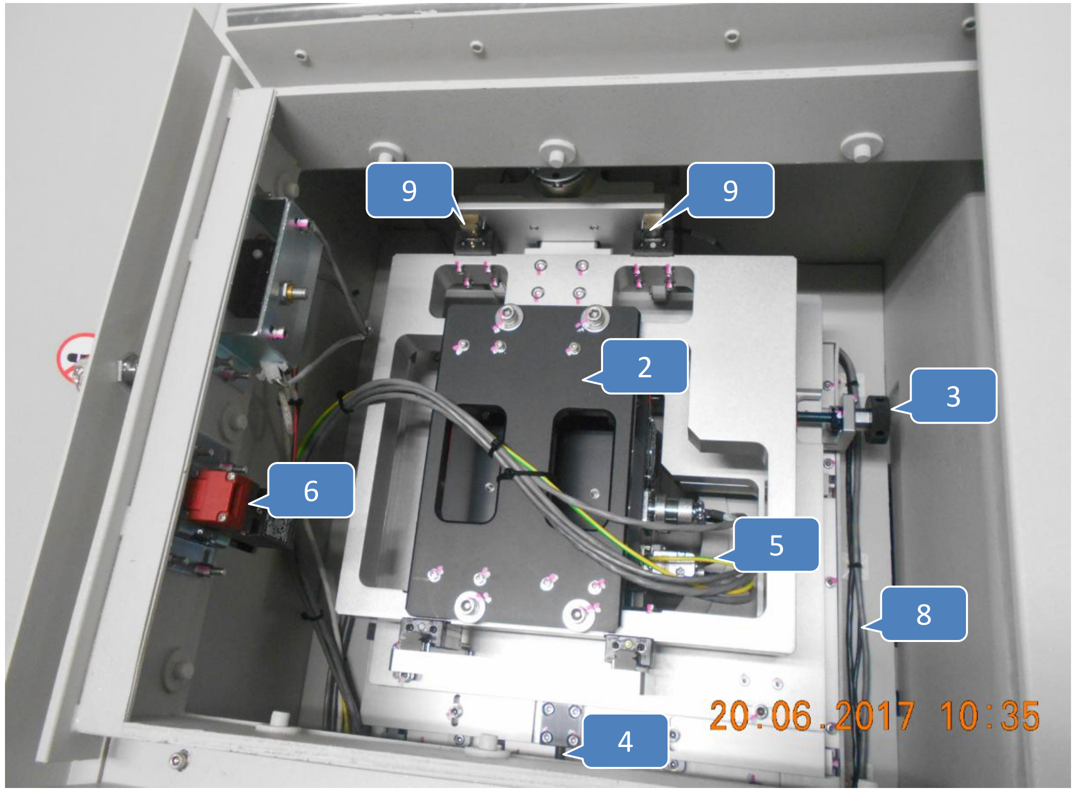
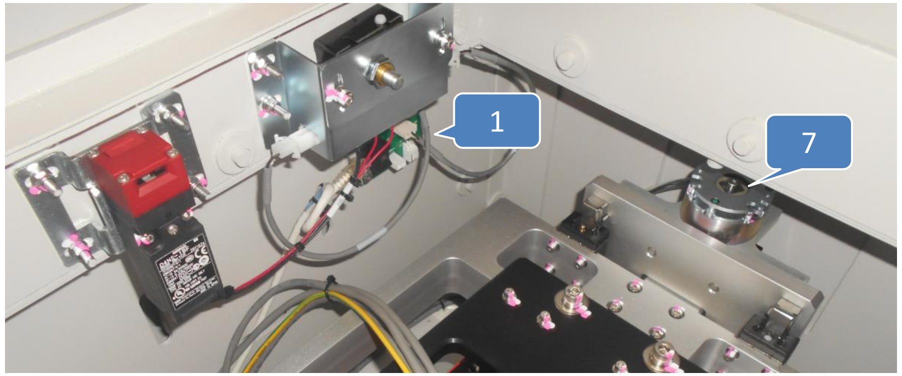
Inner view of Top Access Service Door

1. Break Out Board (BOB 5)
2. X-ray tube with mounting Bracket
3. Tube Y axis position adjustment knob
4. Tube X axis position adjustment knob
5. Connection cables
   - Interlock Chain 1
   - Interlock Chain 2
   - X-Ray tube power input (24vDC)
   - X-Ray tube communication cable (RS232 COM cable)
6. Interlock Switch
7. Tube X height safety bracket
8. Z motor position sensor cables + communication cable
9. Z axis linear guide x 4





System Hardware Overview

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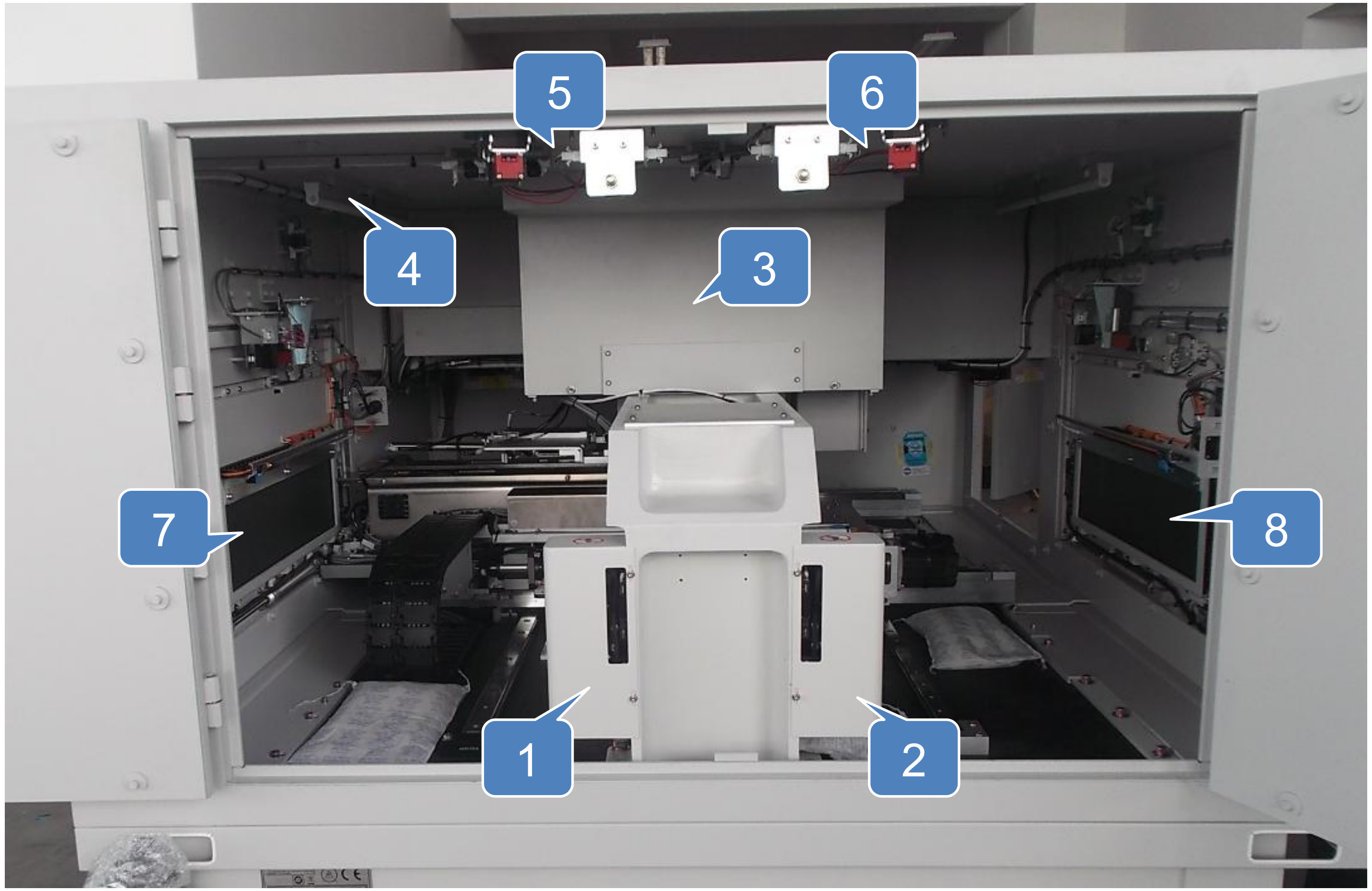
Inner View of System

1. Left Filter Height Module
2. Right Filter Height Module
3. X-Ray Tube wall
4. Service Lamp
5. Left Front Door Interlock Solenoid Switch
6. Right Front Door Interlock Solenoid Switch
7. Left Outer barrier
8. Right Outer barrier



System Hardware Overview

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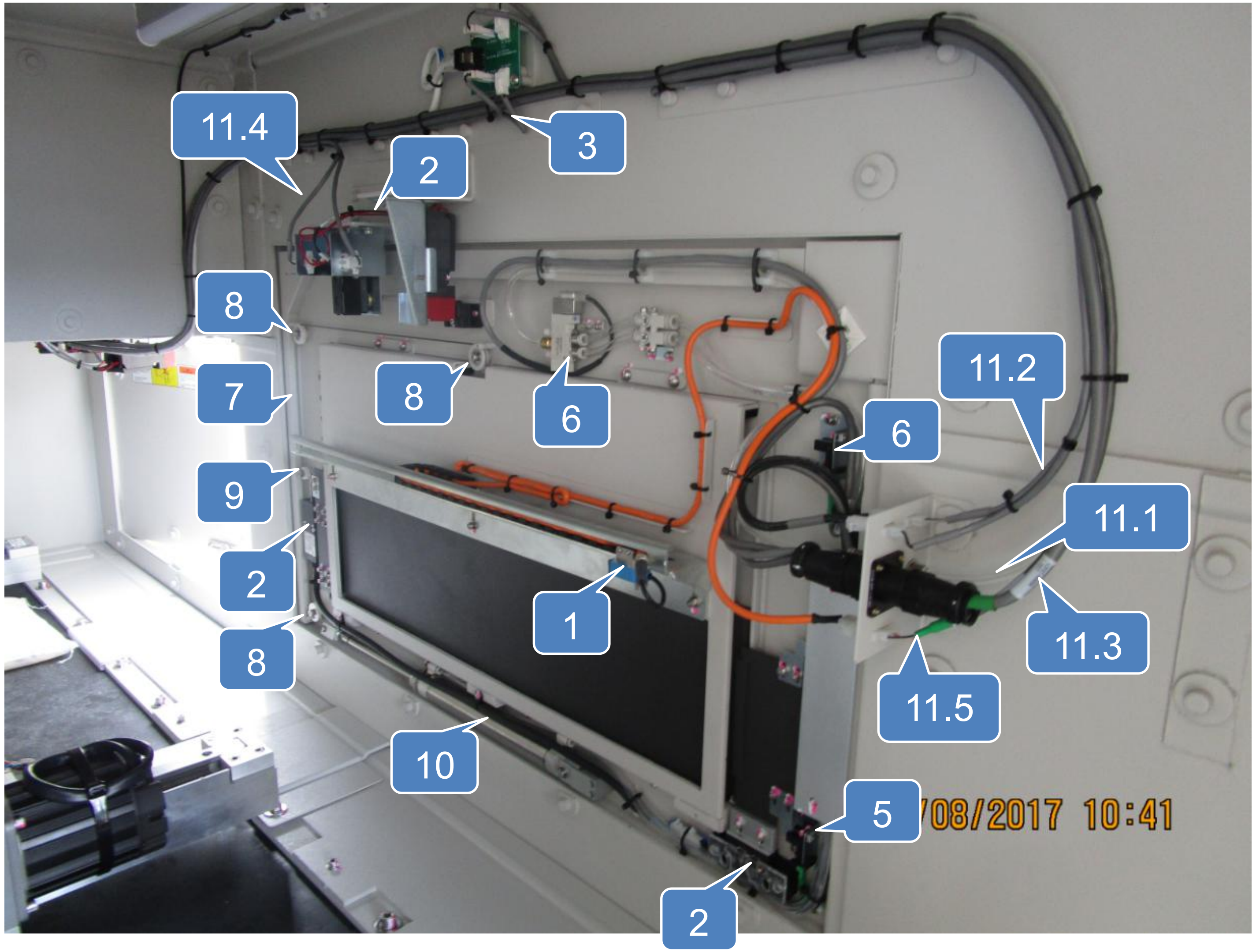
Outer Barrier Layout

1. Panel Clear Sensor (Position adjustable)
2. Interlock Switch Assemble
3. Break Out Board (BOB)
4. Outer Barrier Open Position Sensor
5. Outer Barrier Close Position Sensor
6. Air Valve Controller
7. SAVA cable
8. SAVA cable pulley x 3
9. SAVA Cable Support pulley
10. Outer Barrier Cylinder
11. Connectivity cables
    - 11.1 Air Inlet
    - 11.2 Interlock Chain 1
    - 11.3 Outer Barrier Main Communication cables
    - 11.4 Interlock Chain 2
    - 11.5 Panel Clear Sensor Communication cable



System Hardware Overview

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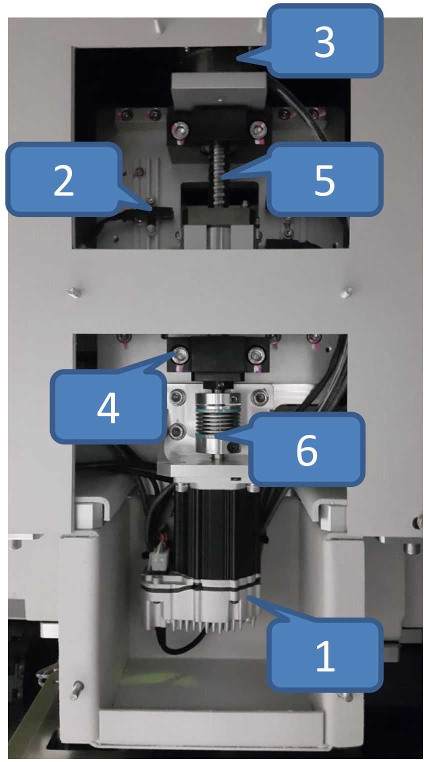
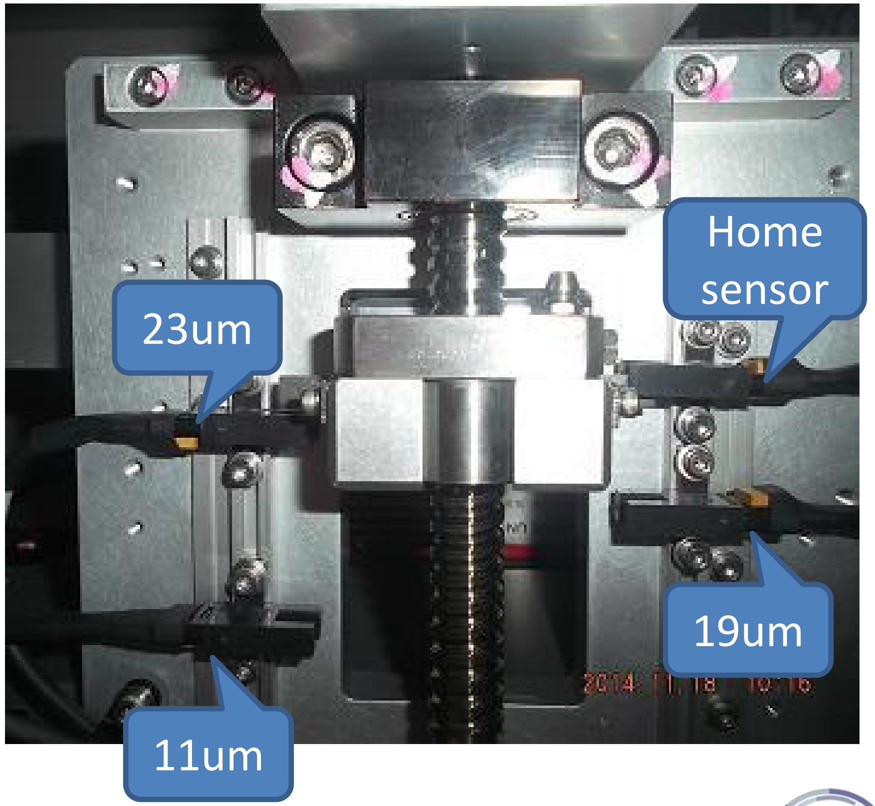
Inner View of Variable Mag Assembly

1. Clear Path Motor (Z-Axis)
2. Position Senor
   - home position
   - 23um
   - 19um
   - 11um
3. Safety Break
4. Bearing Block
5. Ball Screw
6. Coupling





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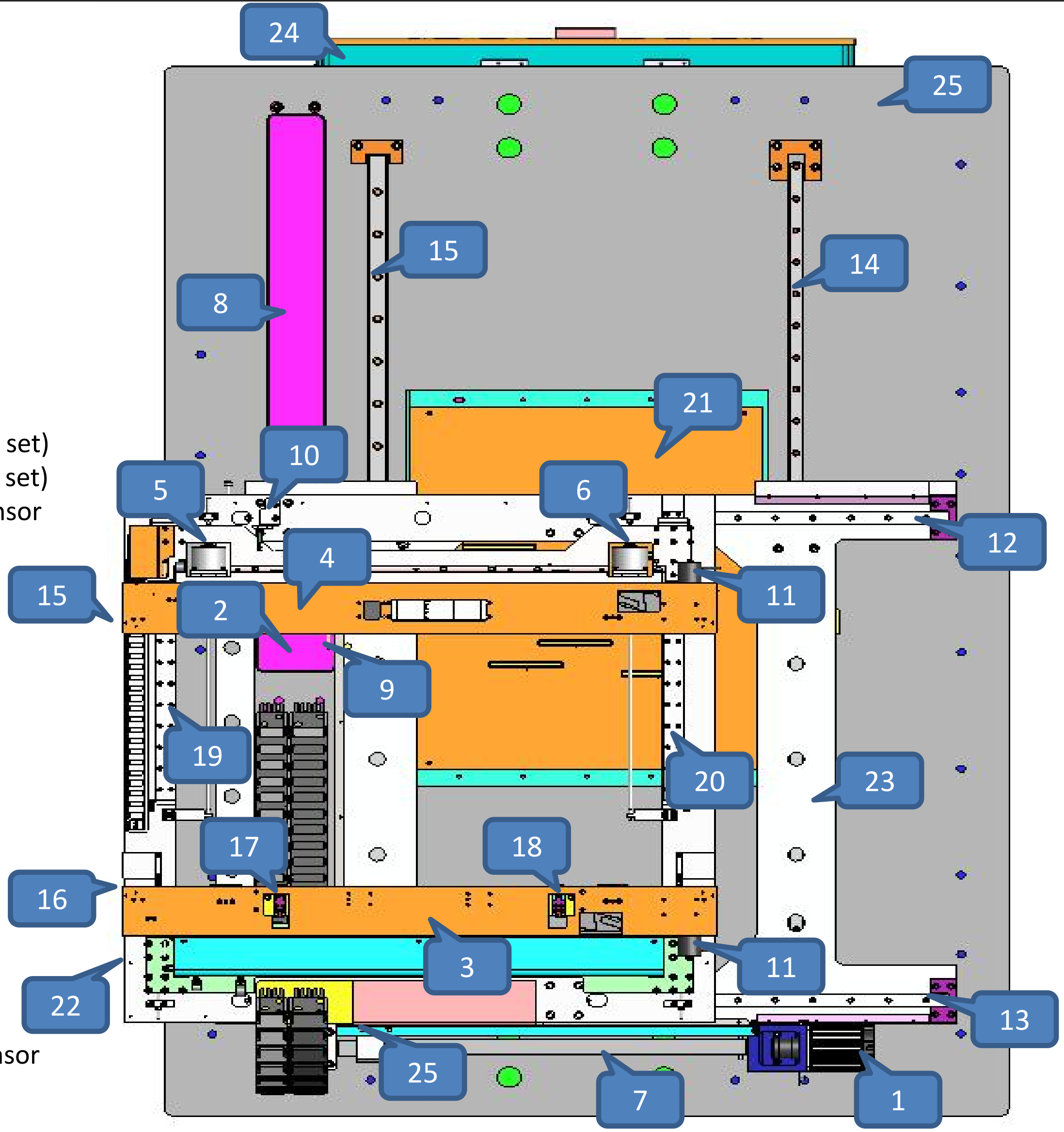
System Hardware Overview

Inner View of XY Table

1. X axis servo motor
2. Y axis servo motor
3. Fix Conveyor Rail
4. Adjustable Conveyor Rail
5. Left AWA motor and lead screw assembly
6. Right AWA motor and lead screw assembly
7. X Axis Ball Screw Assemble (ball screw with fix&support bearing set)
8. Y Axis Ball Screw Assemble (ball screw with fix&support bearing set)
9. Y Axis Encoder read head + Glass Scale + Y axis Home & limit sensor
10. AWA Home Sensor
11. Conveyor Belt DC Motor x 2
12. X Axis Linear Guide x 2 with both end hard stop
13. Y Axis Linear Guide (Wide)
14. Y Axis Linear Guide (Narrow)
15. Fix Conveyor Rail Panel Clamp and Sensor
16. Adjustable Conveyor Rail Panel Clamp and Sensor
17. Left Mechanic PIP Stopper
18. Right Mechanic PIP Stopper
19. Left AWA motor linear Guide
20. Right AWA motor linear Guide
21. Camera Field Stop Assemble
22. X Axis Base Plate
23. Y Axis Base Plate
24. MCA/SSCA Box
25. X axis Encoder read head + Glass Scale + Y axis Home & limit sensor
26. Granite base plate



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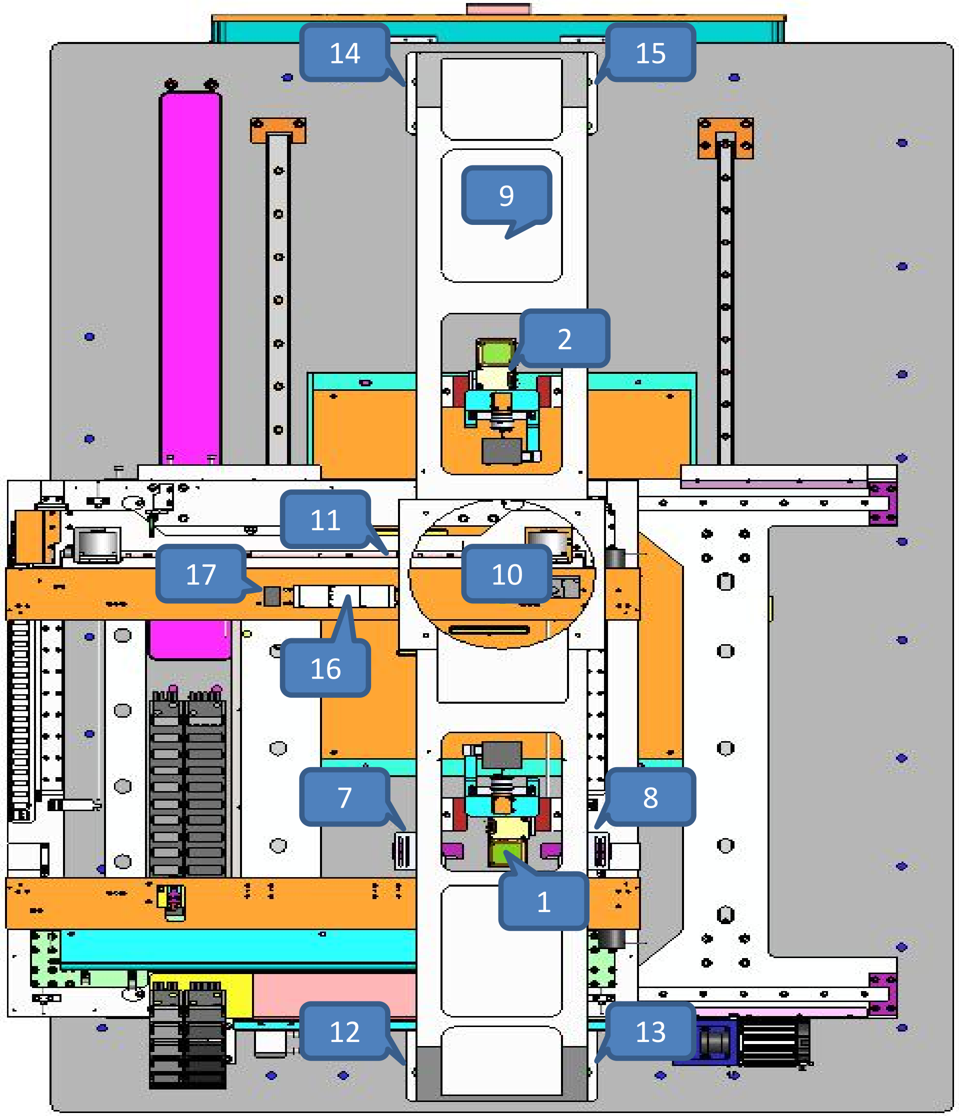
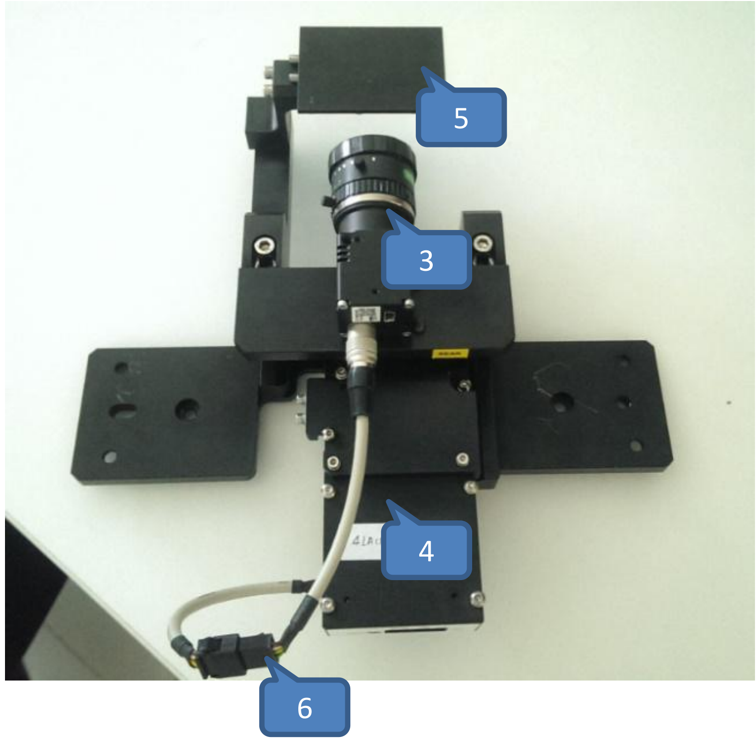
System Hardware Overview

Inner View of XY table

1. Front PSP Module
2. Rear PSP Module
3. Optical Camera and Lens
4. Projector
   (Configure Switch IP + IP address)
5. Mirror (fix Angle)
6. Trigger cable
7. Right PIP Slow Sensor
8. Left PIP Slow Sensor
9. Cast Iron Beam
10. X-Ray Tube
11. Inner Barrier Assemble
12. Left Filter Height Sensor Module
13. Right Filter Height Sensor Module
14. Left Filter Height Sensor Reflector
15. Right Filter Height Sensor Reflector
16. PSP On-rail Jig
17. PSP Grey Card





System Hardware Overview

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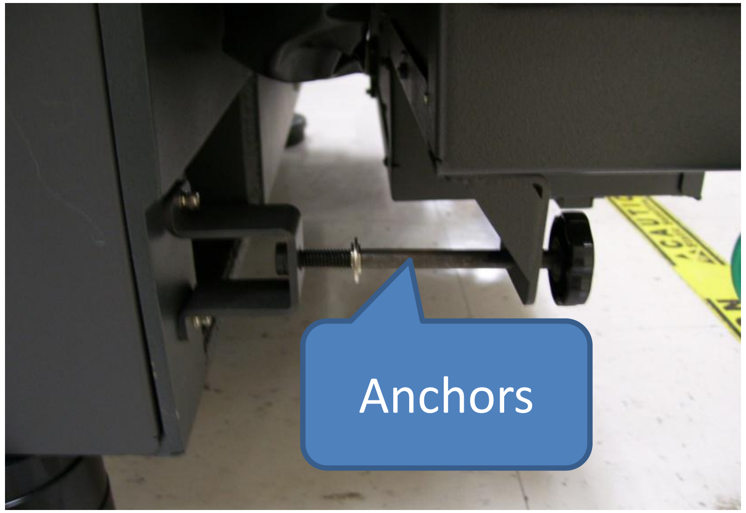
Rear of System

- ➢SSCA Normal Position including ‘anchors’
- ➢SSCA Service Position including support



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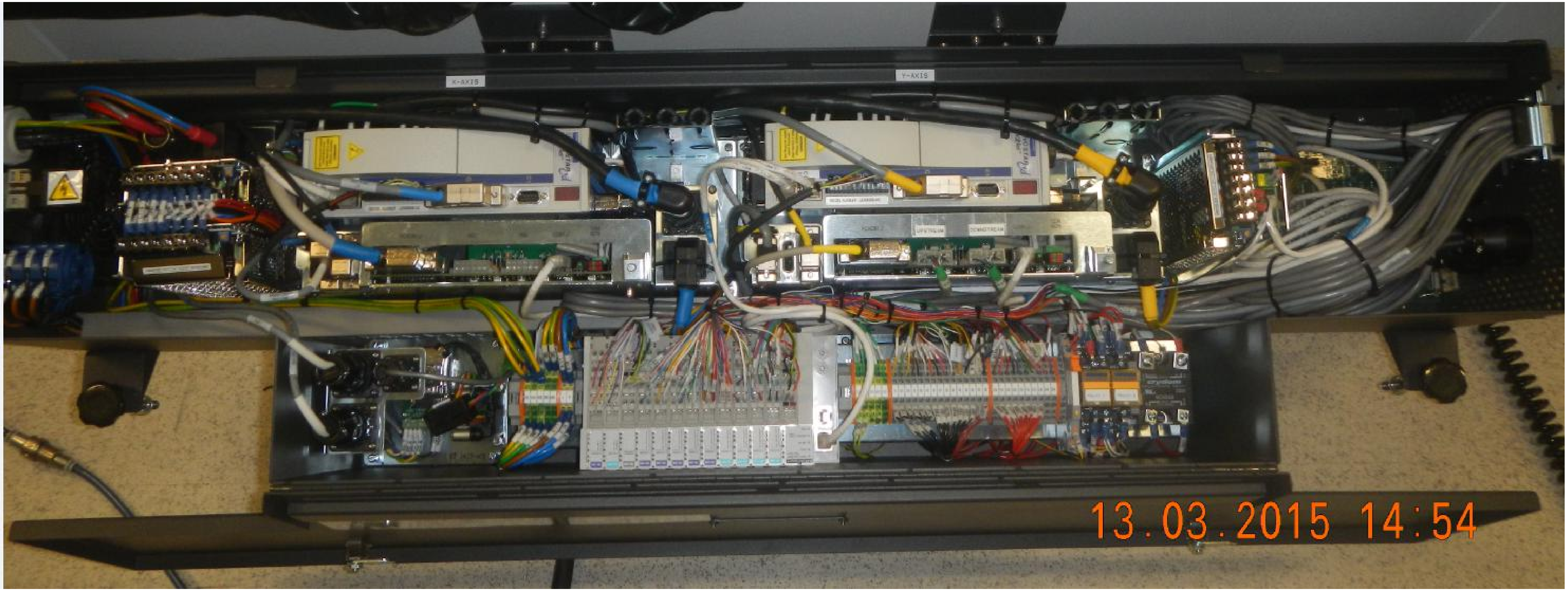
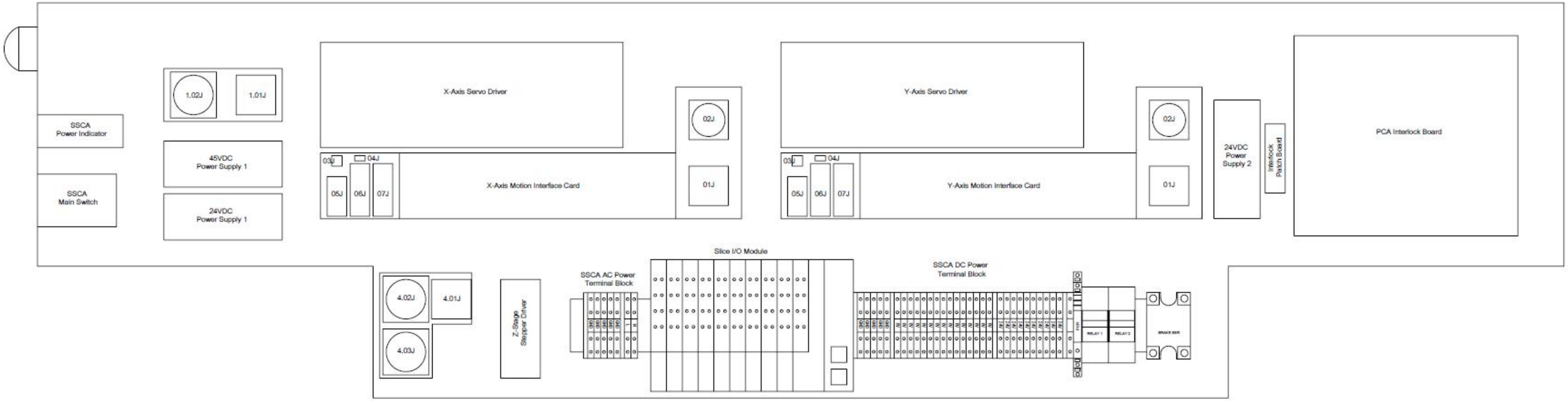
System Hardware Overview - SSCA/MCA Box



System Hardware Overview - SSCA/MCA Box Layout

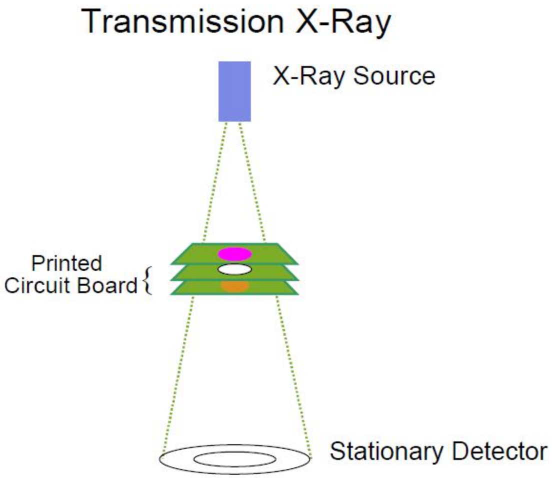


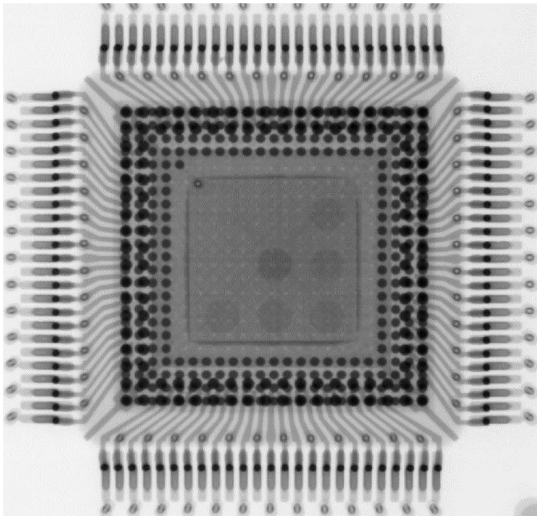
Inspection Theory - Overview

Automated Board Inspection

- X-ray beam passes through the printed circuit board.
- Dense metal absorb more of the X-ray’s energy than less dense material.
- The X-ray beam reaches the “Detector” (X-ray Camera).
- The “Detector” converts the X-ray energy into a gray scale image of a solder joint.
- The “Detector” then sends the gray scale image to the Imaging Reconstruction Processor (IRP).
- IRP collect the images, reconstruct the images to 3D Reconstructed Images and send the images to the computer.
- Computer identifies and measures specific pieces of the joint’s image.

Transmission X-ray (2D Image)





Transmission X-ray:

- The X-ray mean that most people are familiar with is the kind used by a doctor or dentist. These are "transmission" X-ray and regardless of the layering of the objects being x-ray produced an image of all features.

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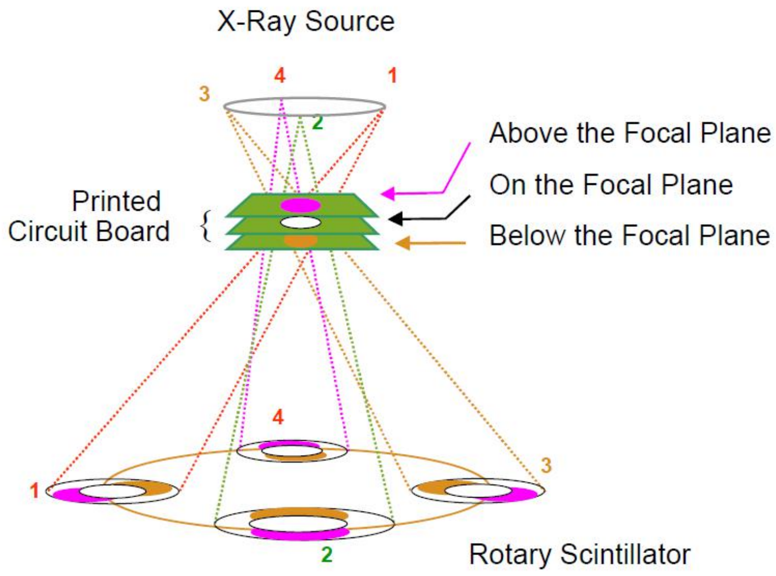
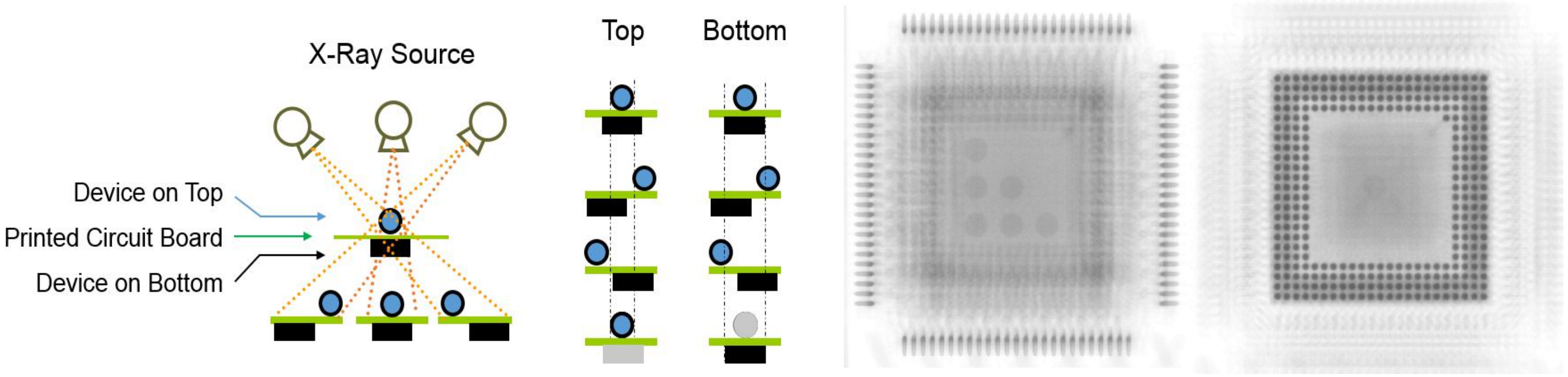
Laminography (5DX)



Laminography X-ray Beam:

- The X-ray mean is rotating. Deflection coils in the tube cause the photon stream to pass through the panel at a rotated angle.
- Agilent 5DX uses X-ray Laminography.

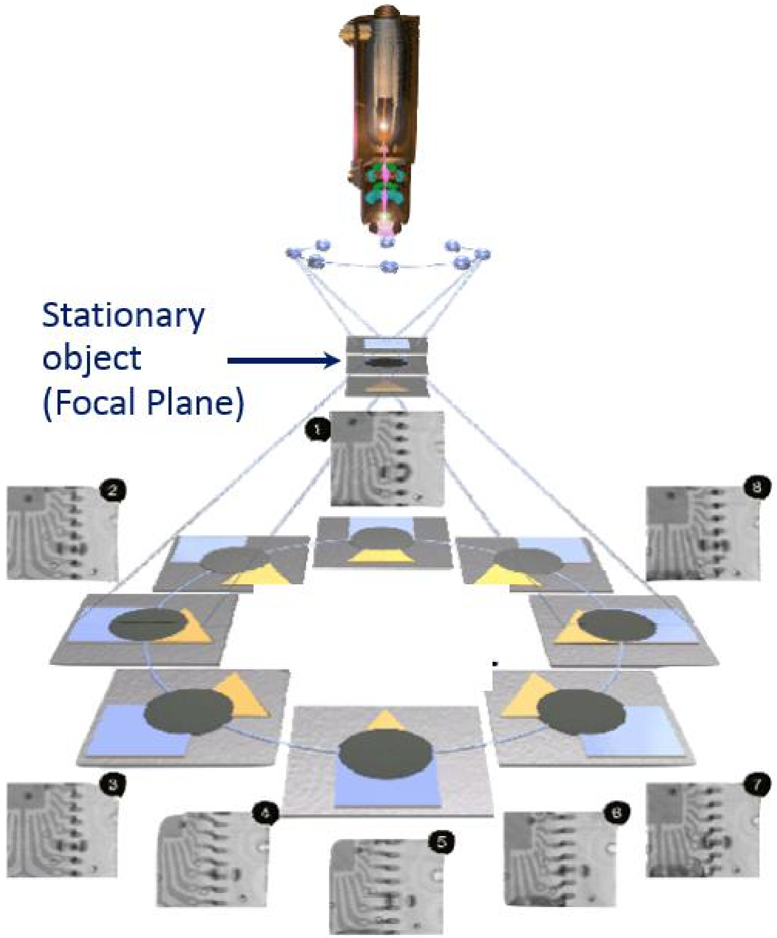
Digital Tomosynthesis (V810)



Digital Tomosynthesis X-ray Image Reconstruction:

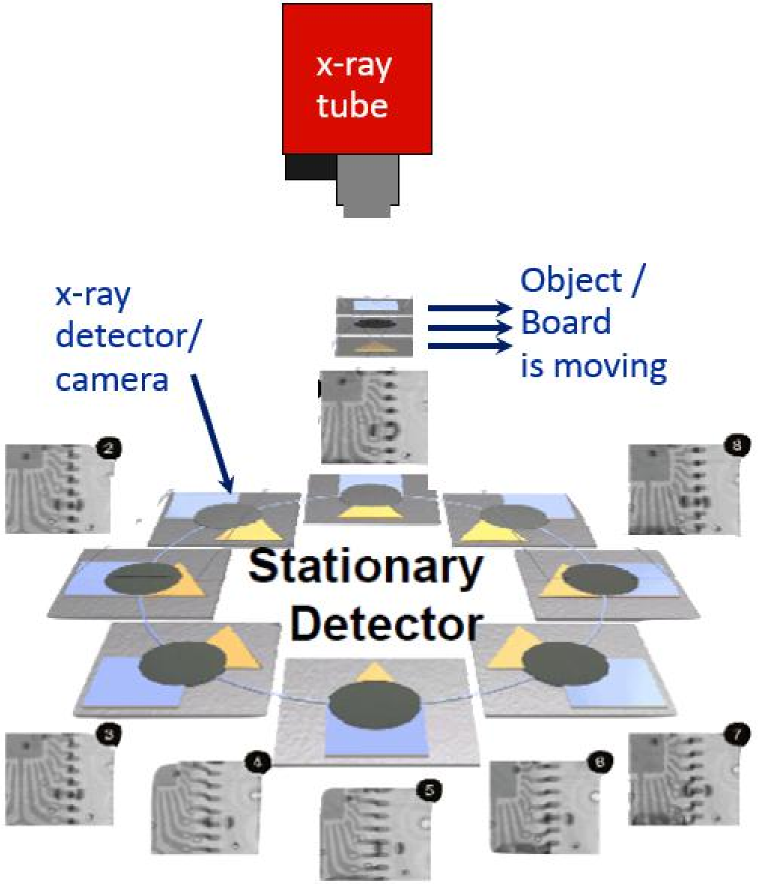
- Uses Transmission X-ray, and capture the images in different angle.
- The images from different angle is then combined to reconstruct 3D images.
- ViTrox V810 AXI system uses Digital Tomosynthesis X-ray Image Reconstruction Technology.

Example – 5DX Laminography



- Laminography required oblique angle images for 3D reconstruction.
- Relative motion of the source, detector, and object under inspection generates a cross-sectional image by smearing into the background, all information not located on a predetermined plane-of-focus (POF).
- Because only 1 areal mode camera is used, multiple angle images can only being obtained by moving both x-ray spot and detector.

Example - Tomosynthesis



- As opposed to 5DX's Laminography, object / board is moving in Tomosynthesis during image acquisition while x-ray spot and detector (camera) are not.
- To obtain different angle images, multiple TDI cameras are installed at different angle with respect to the object

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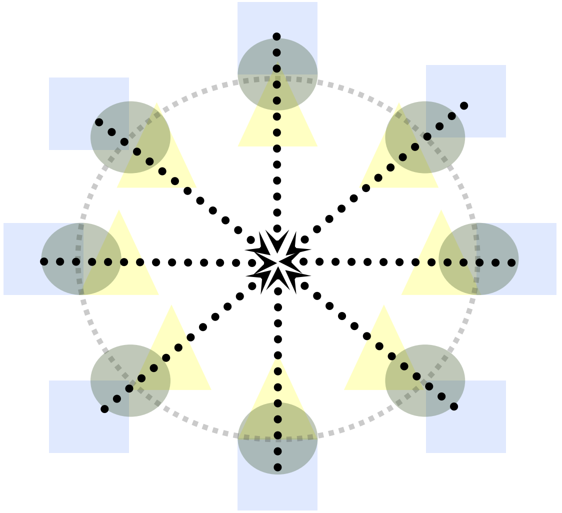
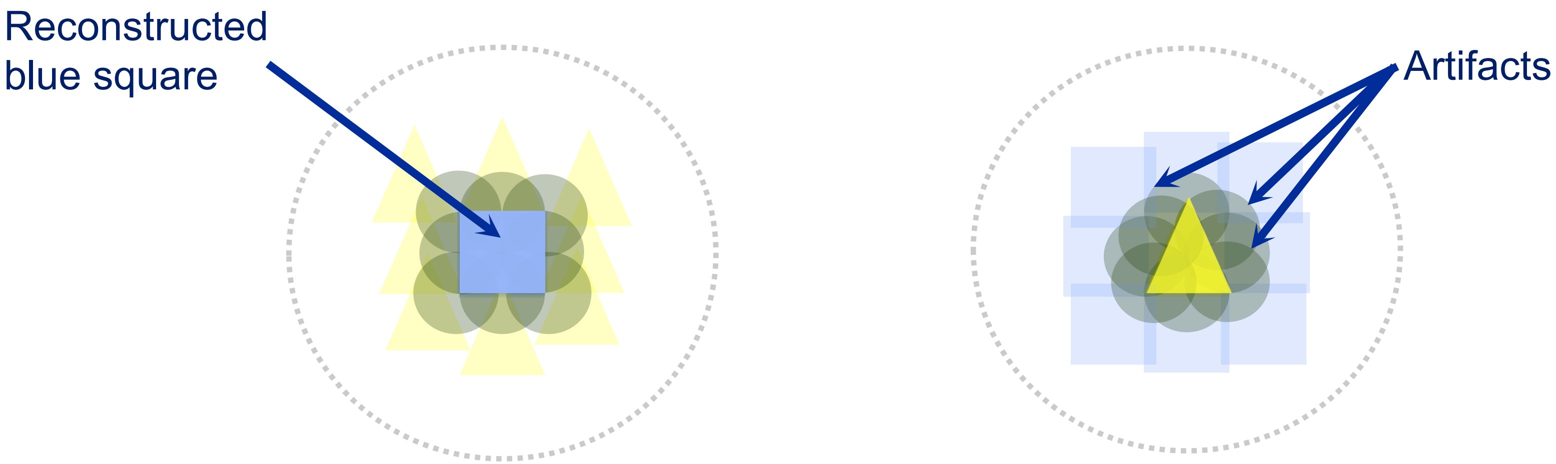
Example – Image Reconstruction Process





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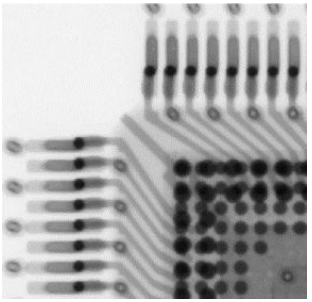
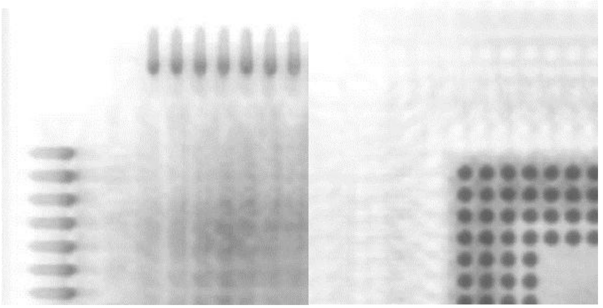
Comparing 2D and 3D Images





Comparing 2D and 3D images:

- Image on the top is 2D image
- Image on the bottom are 3D image
- All images are from the same PCB, same region.

Notice:

- 2D image contains information from the top and bottom of the panel
- 3D image contains information from only one side of the panel
- Edges in 2D image are crisp
- Edges in 3D image are fuzzy
- If there are voids on the solder ball
  - 3D image can detect but 2D image don’t

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Digital Reconstruction

- Uses industry standard techniques for oblique image reconstruction.
- Allows for arbitrary number of in focus planes from a single acquisition sequence.
- V810 system uses multiple shift and combine methodologies
    - Different methods for defect image optimization: shift+average, shift+ordered statistics, etc
    - Multiple standard statistical noise reduction techniques, such as, averaging, double correlated sampling, etc.

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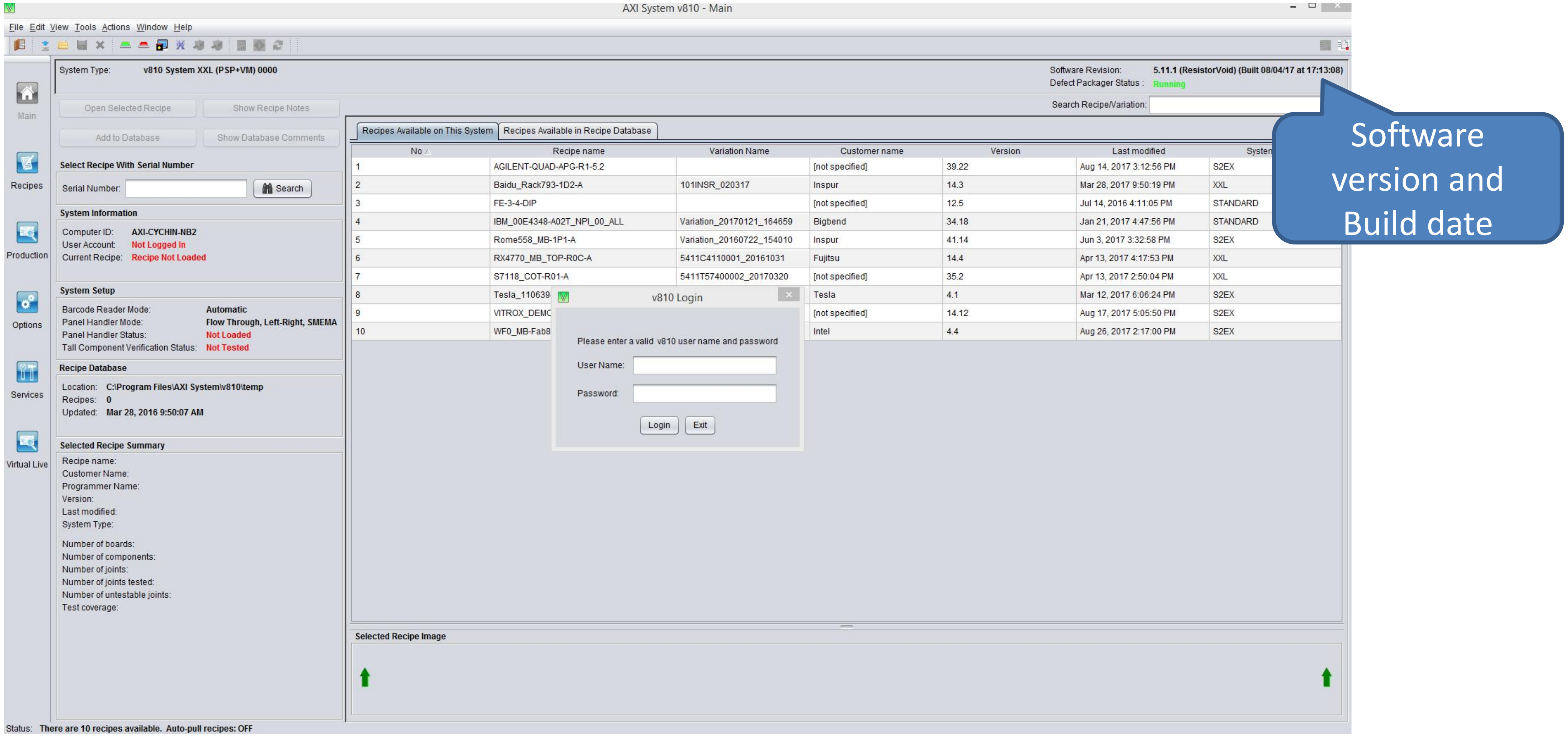
System GUI -Main Page



- V810 GUI Main page, shown “Recipes” list and SW version information
- Only able to proceed other tab once log in assign password
- Enable or Disable Defect Packager (Icon on top of Software version)

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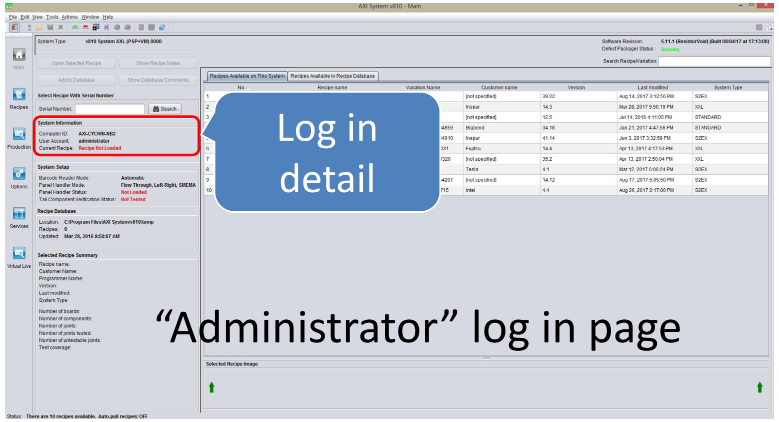
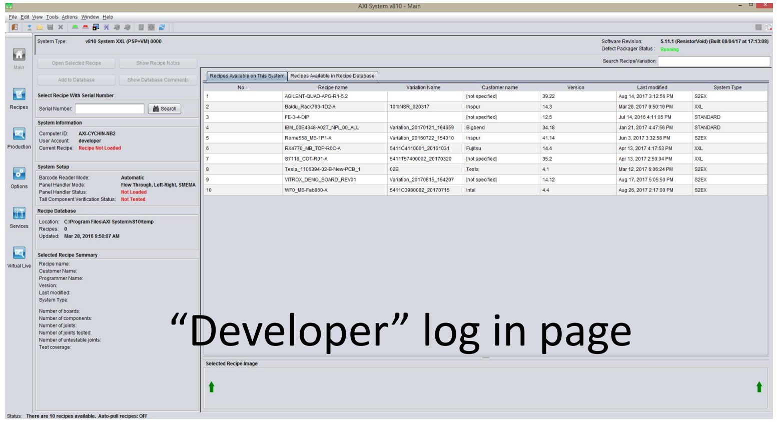
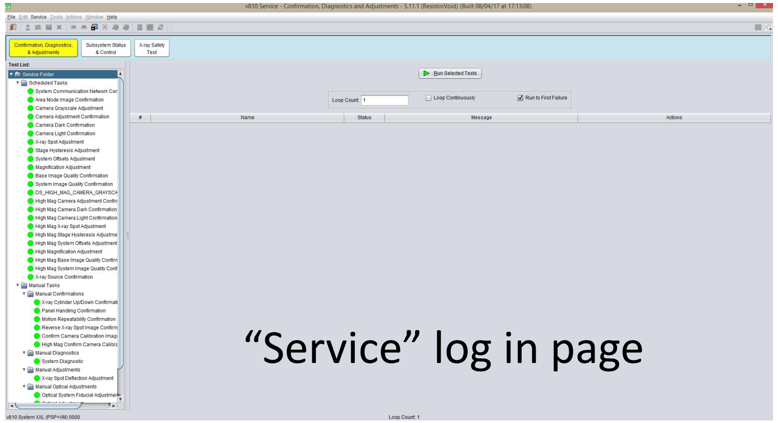
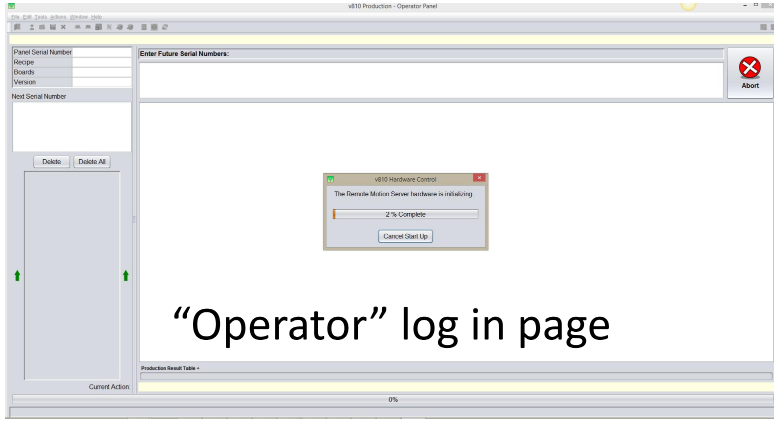
System GUI - Main Page - Log In Password









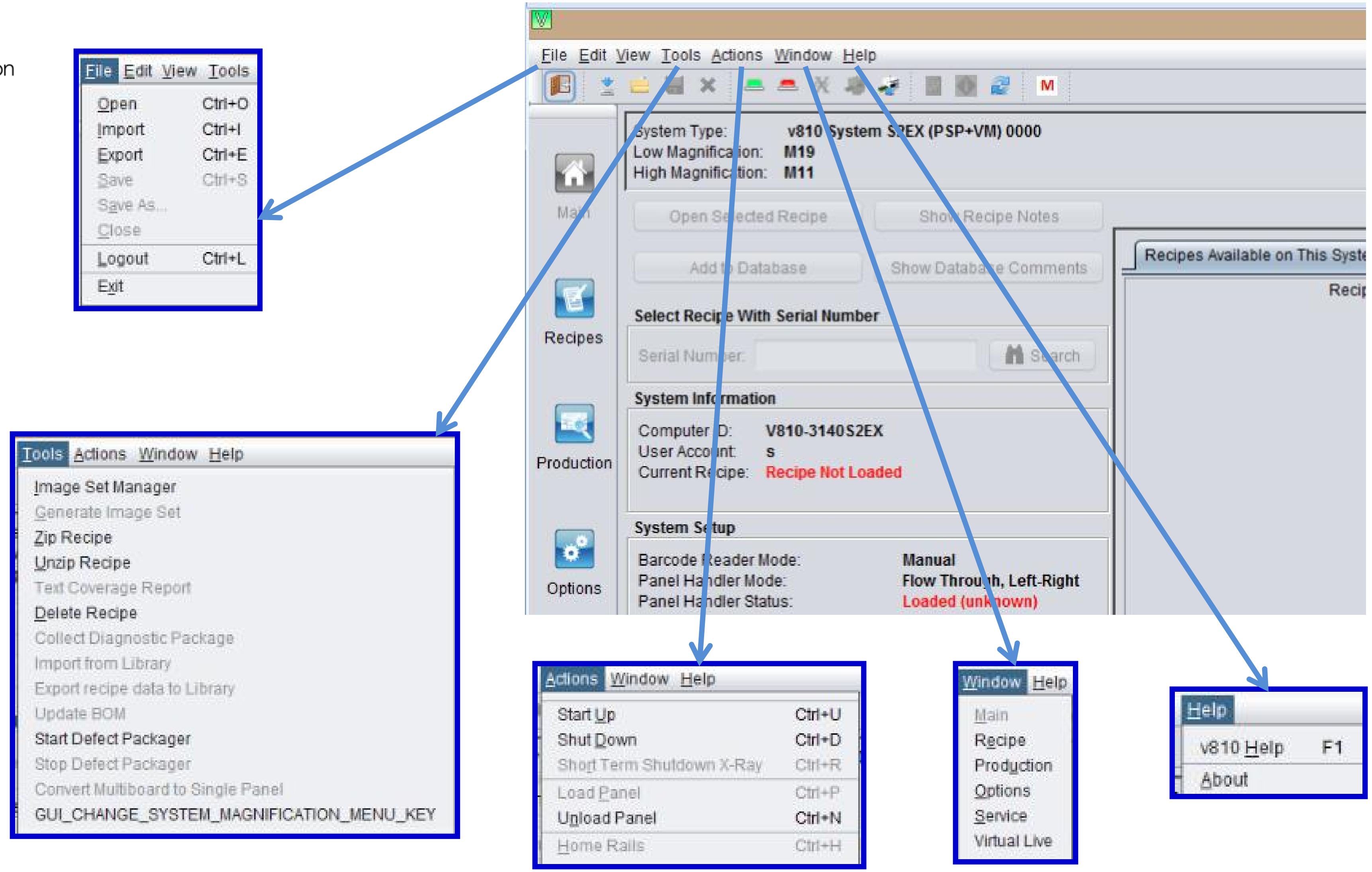
- **Window 8 (Main controller),** default User name, **Administrator**, Password, **Please!**
- **Window 7 (RMS/Atom PC)**, default User name, **Administrator**, Password, **Please!**
- **V810 GUI**, 4 default User Names group create with same password but difference authority

  - User Name, **Administrator/Developer/Service/Operator**
  - Password: **nebulae**

- User can create own account with desire authority at Option > Customer Option tab

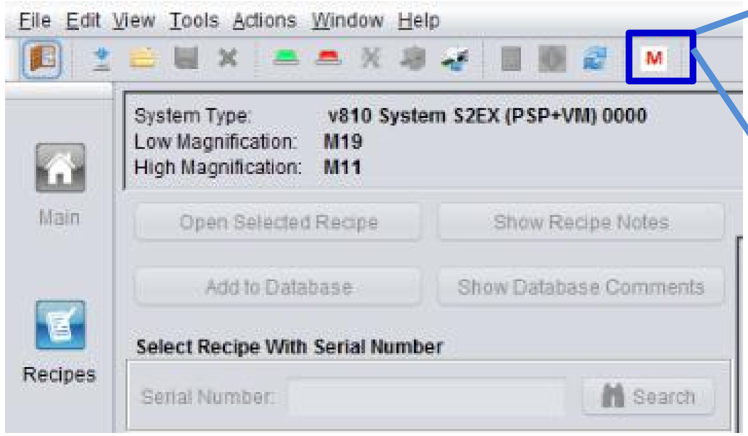
System GUI - Main Page - Interface

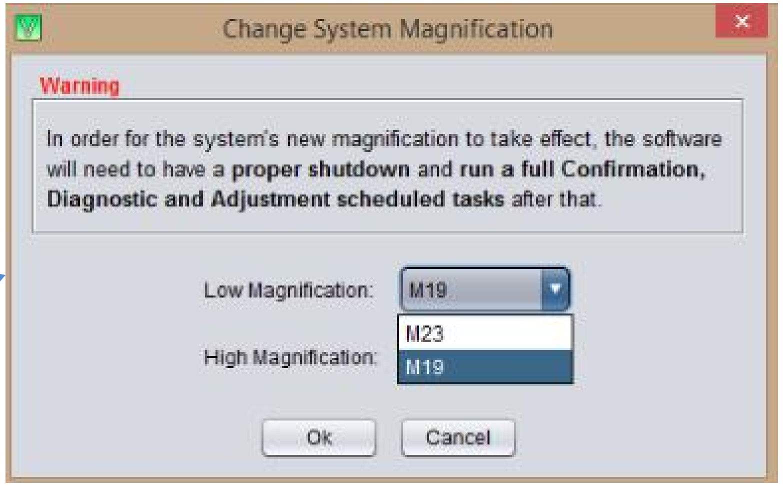
Automated Board Inspection



ABI System GUI - Main Page - Low Magnification Switching

Automated Board Inspection





Low Mag Consist 2 type of magnification

High Mag Consist single type of magnification

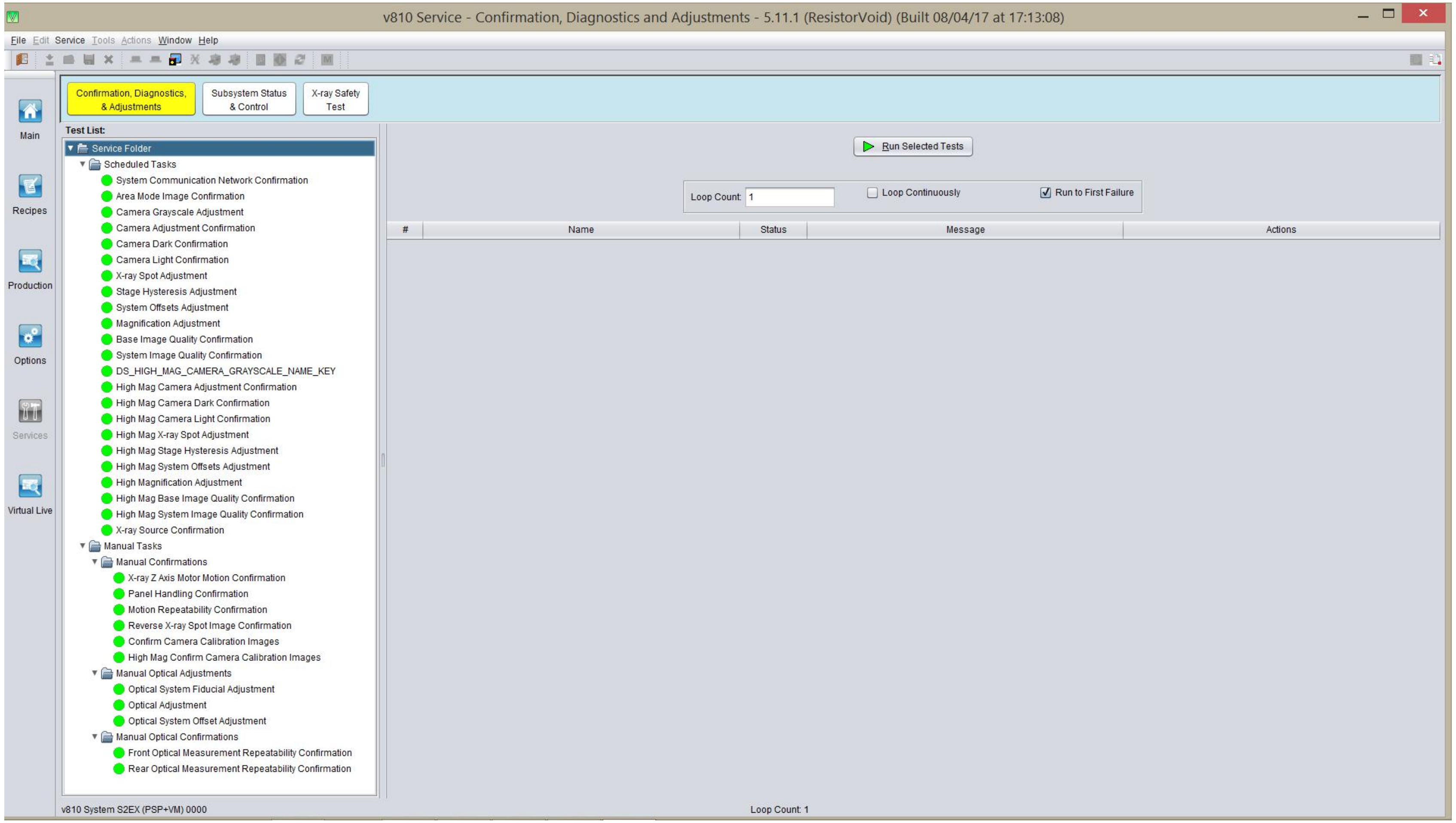
Remarks :

Any changes on magnification on this “M” tab , software GUI will required to “ Refresh”

ViTrox

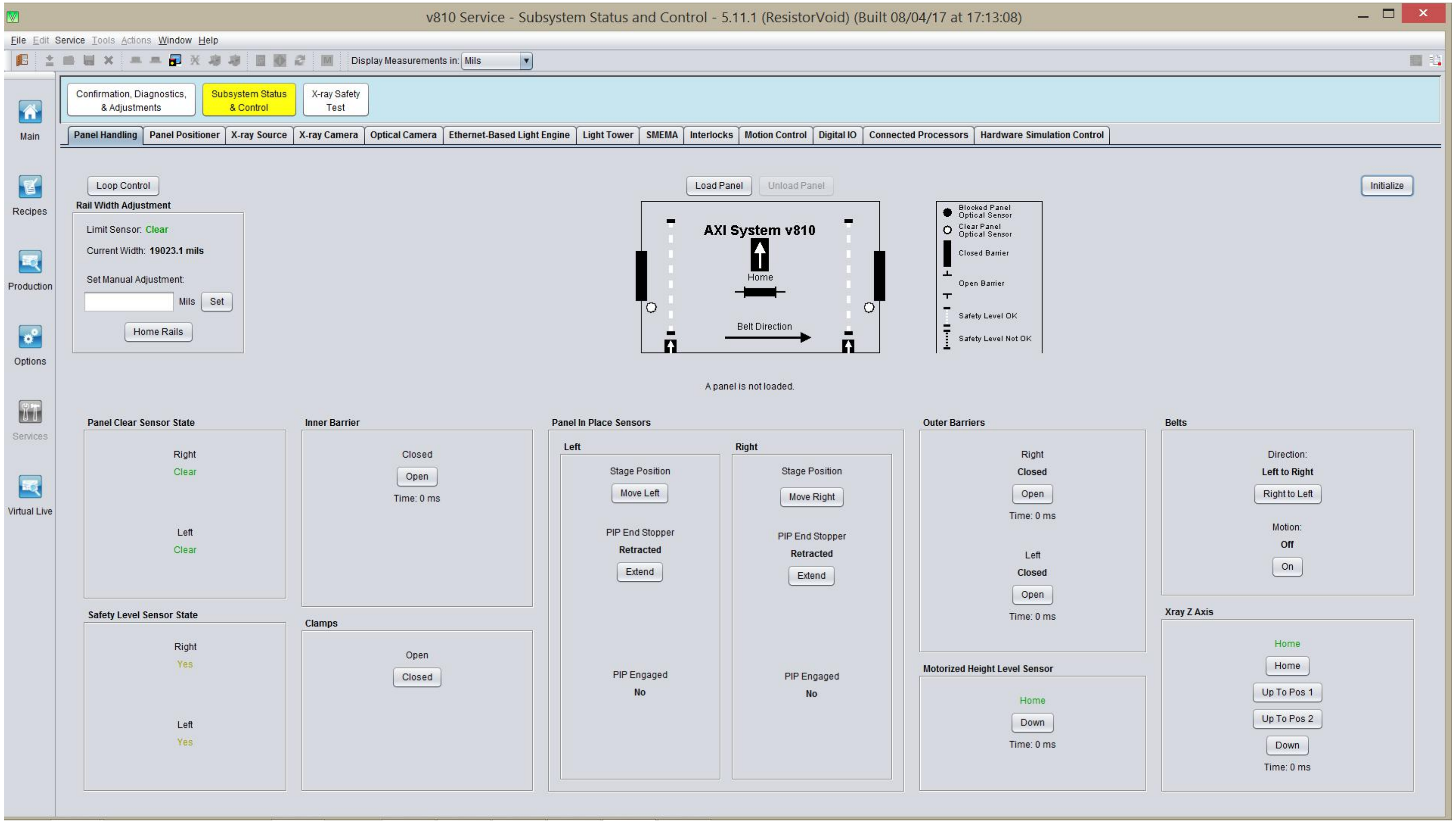
Automated Board Inspection

System GUI - Service tab - Confirmation Diagnostics and Adjustment tab



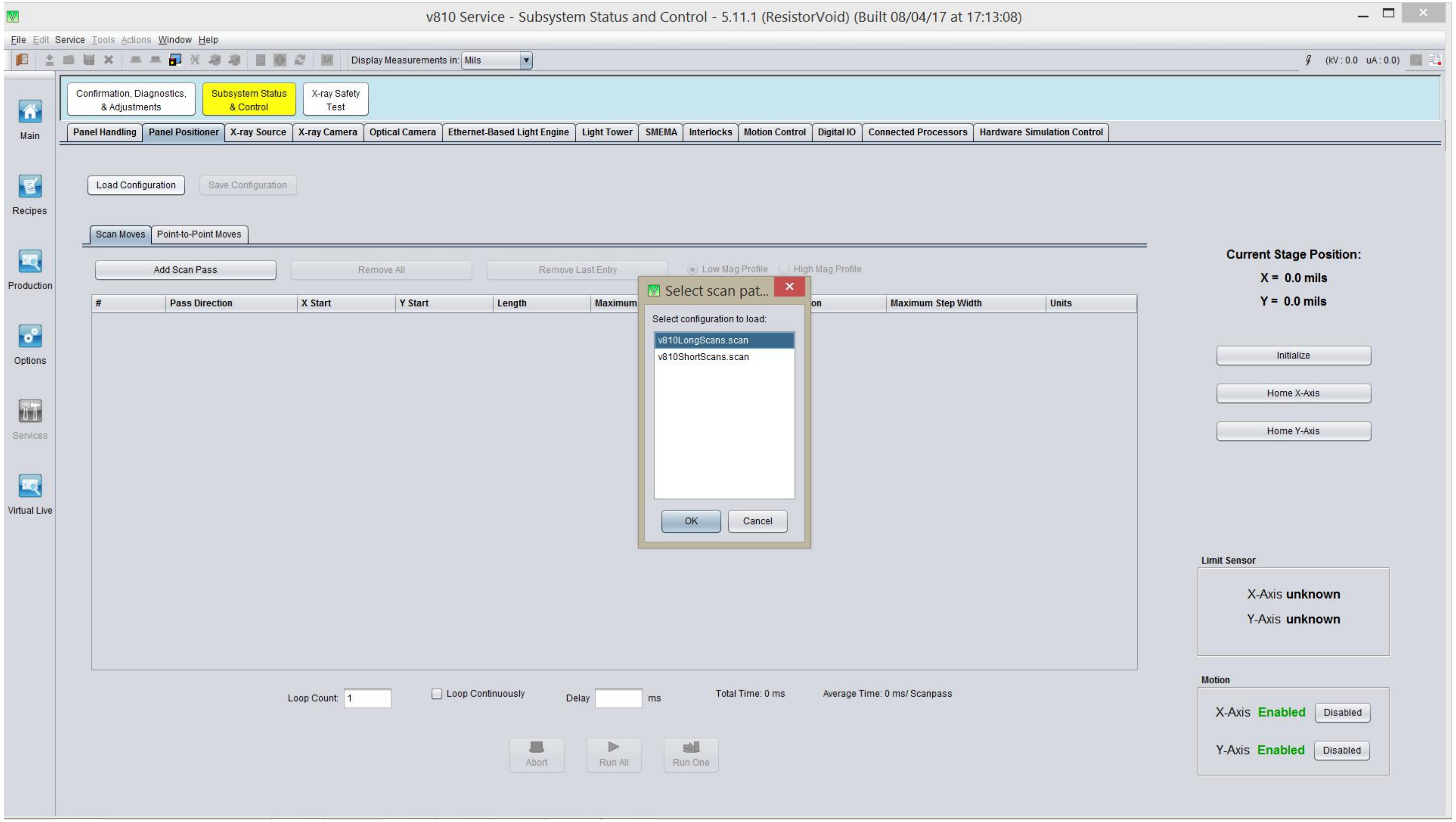
- Confirmation and adjustment before perform board inspection
- An automatic periodic checking will be carried out at an interval time to make sure the system condition is within intended specification. And if found not within intended specification, the software will adjust its calculation to make sure the inspection performance is consistent with previous inspections. Until a point that calculation adjustment is impossible, the machine will prompt message stating the machine needs to be serviced.

System GUI - Service tab - Subsystem Status & Control Page - Panel Handling tab



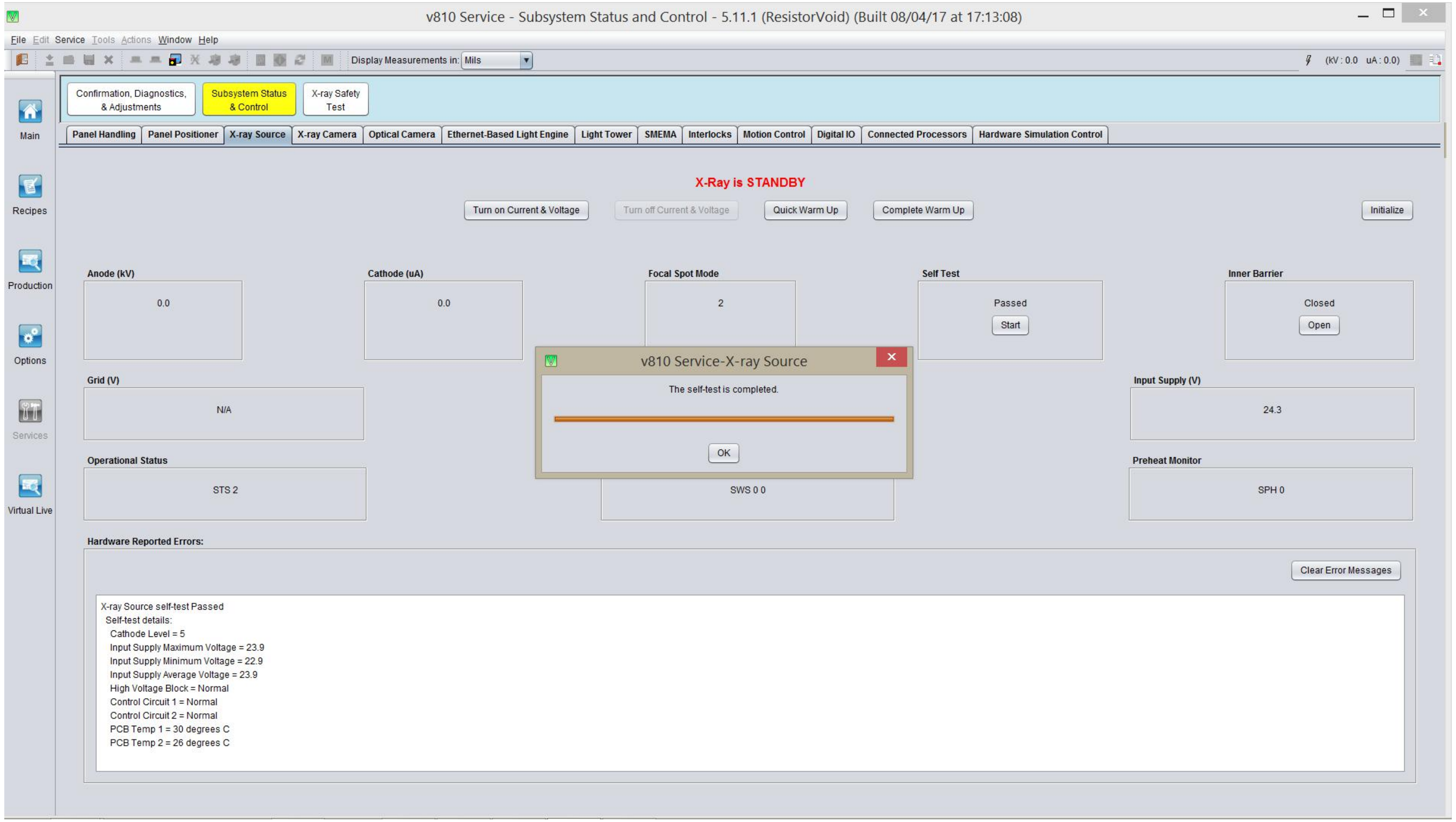
- Unable to initialize if Panel Clear Sensor input value incorrect or interlock break
- Same function as DIO tab with simply view (#manually toggle output bit and feedback the input value)
- Show movement speed for inner barrier/outer barrier/dual mag
- Home AWA and manually set width as per wish
- Manual load panel with set width

System GUI - Service tab - Subsystem Status & Control Page - Panel Positional tab



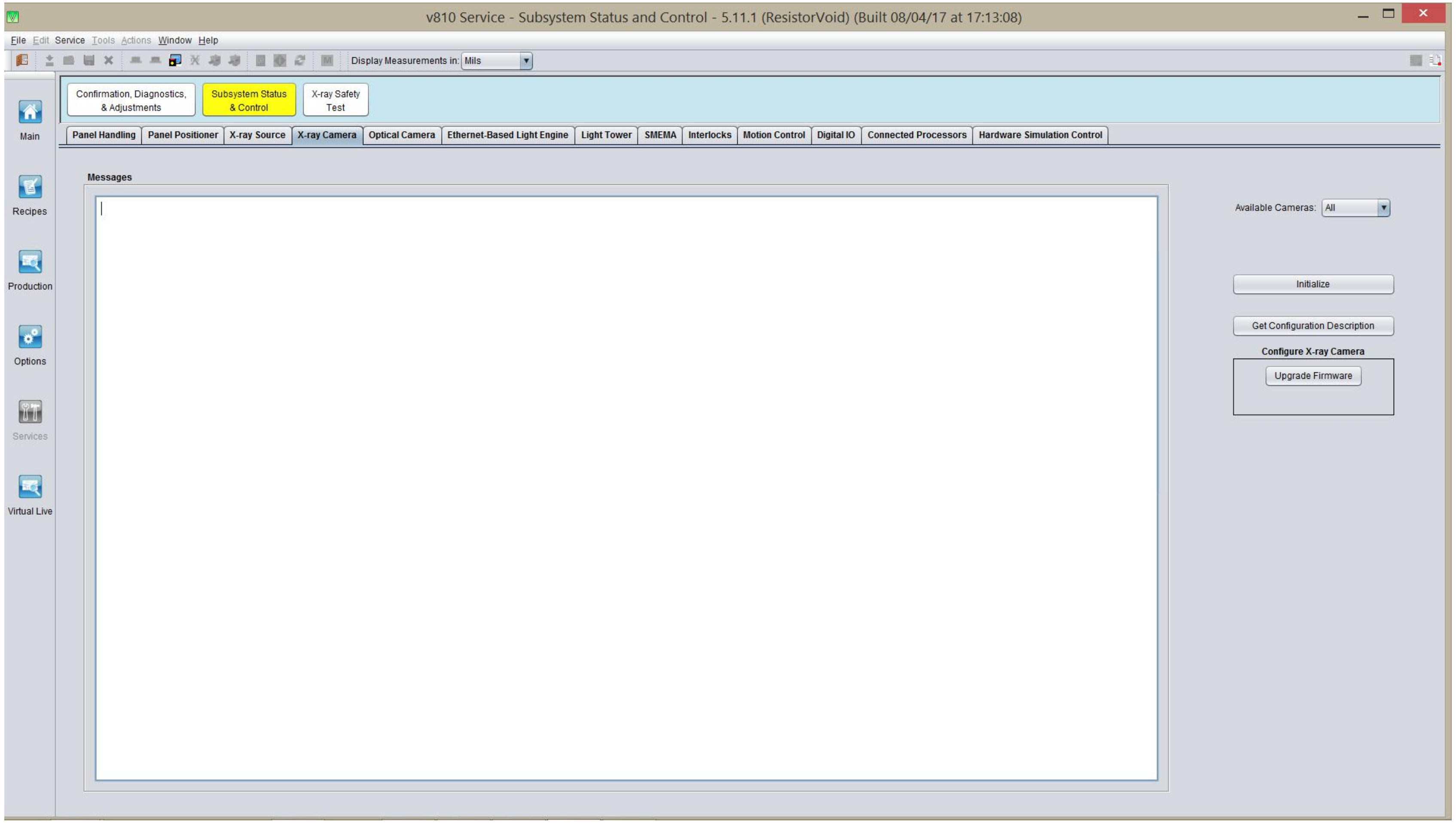
- Unable initialize if Interlock break or motion hardware failure
- Motion repeat-ability checking
- Current stage position information
- Allow individual home/ move stage position with given command

System GUI - Service tab - Subsystem Status & Control Page - X-Ray Source tab



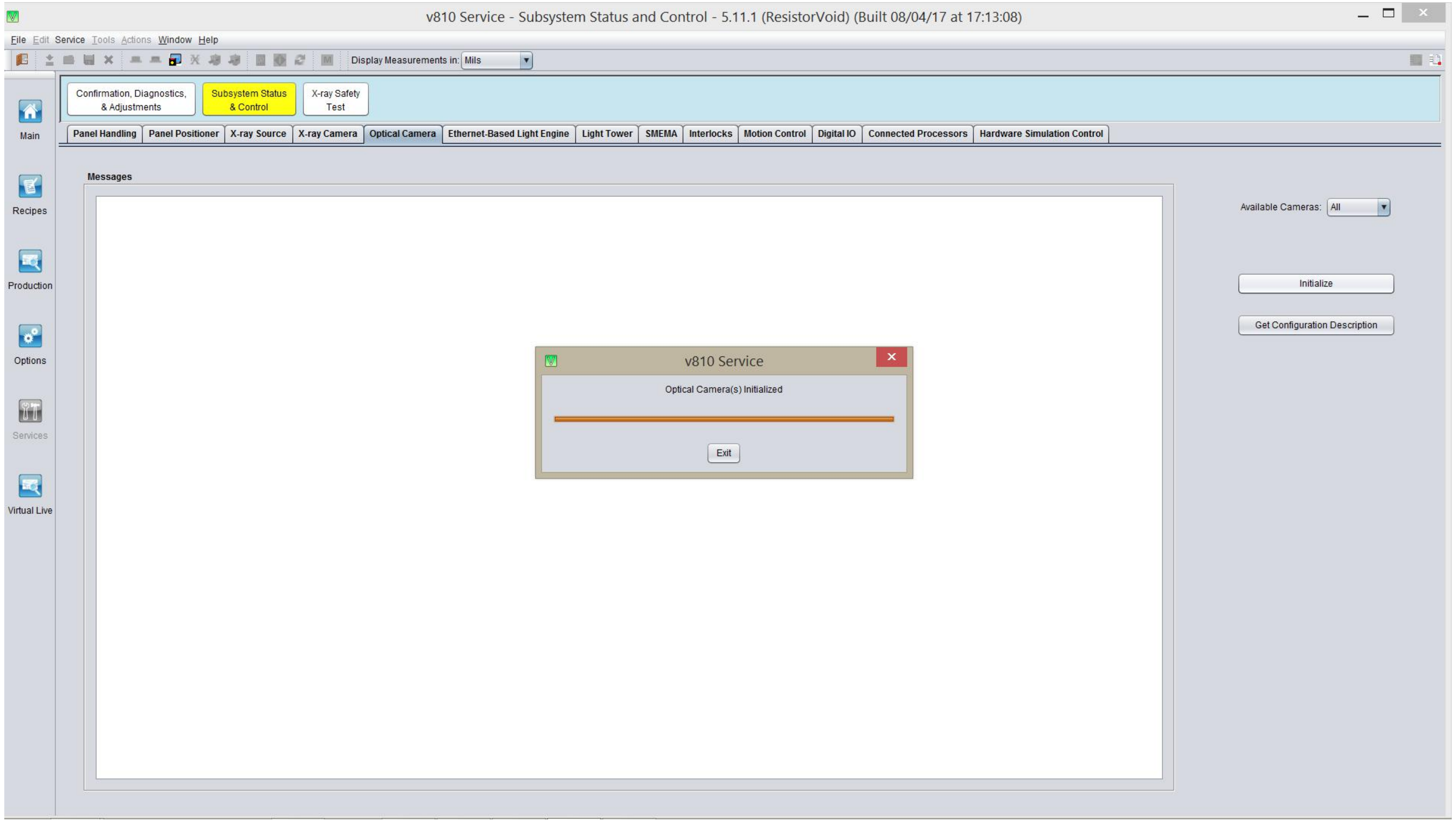
- On/Off tube power
- Perform tube warm up
- Perform tube self test and provide tube current status
- Unable initialize if interlock break or tube not in ready status.

System GUI - Service tab - Subsystem Status & Control Page - X-Ray Camera tab



- Camera firmware upgrade base on camera version
  # If change or swap AXI X-Ray camera, the Camera upgrade firmware MUST be performed
- Check camera communication status with system (all cameras or individual)
- Unable Initialize or perform firmware upgrade if connection error

System GUI - Service tab - Subsystem Status & Control Page - Optical Camera tab



- Communication status verification for PSP optical camera with main server
- Unable initialize if hard ware failure.
  (# Possible Root cause; USB cable not seat firmly or optical camera power lost)

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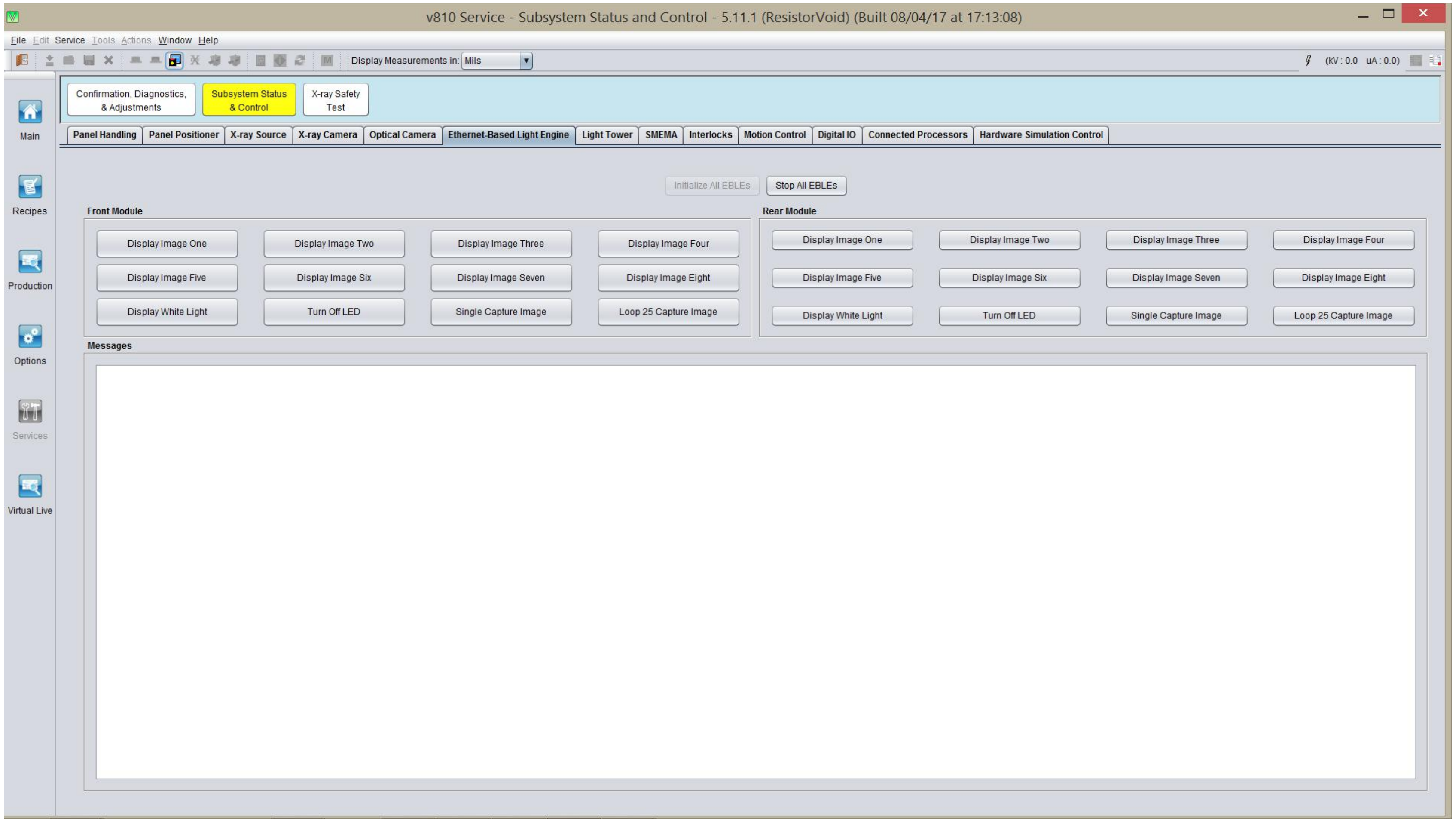
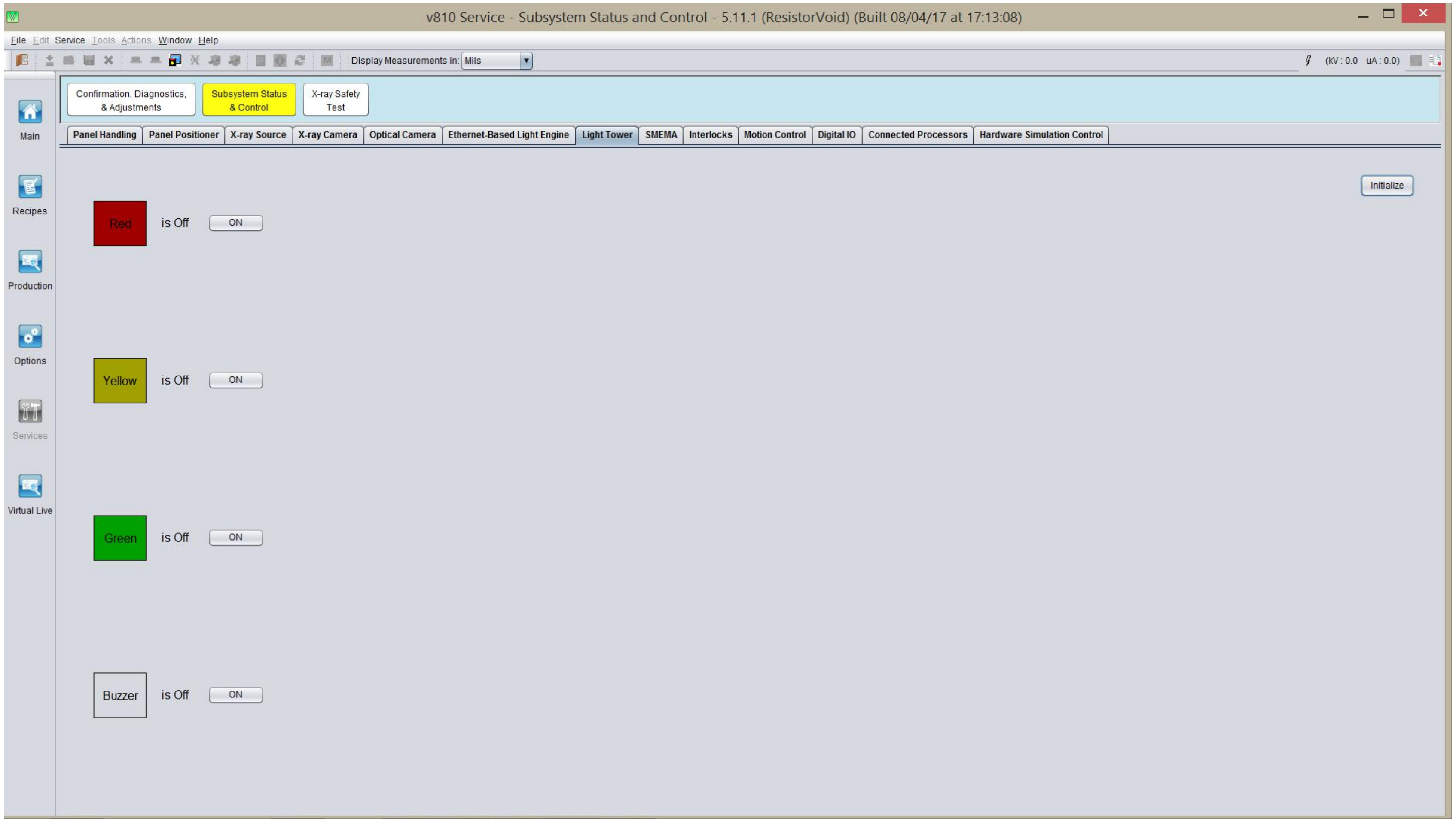
System GUI - Service tab - Subsystem Status & Control Page - EBLE tab



- PSP 2 projector main tab
- Manually trigger project with selected fringe pattern for focusing verification
- Unable initialize if power lost at projector or trigger cable connection issue

ABI
Automated Board Inspection

System GUI - Service tab - Subsystem Status & Control Page - Light Tower tab



- Manual trigger Tower light functional for verification purpose

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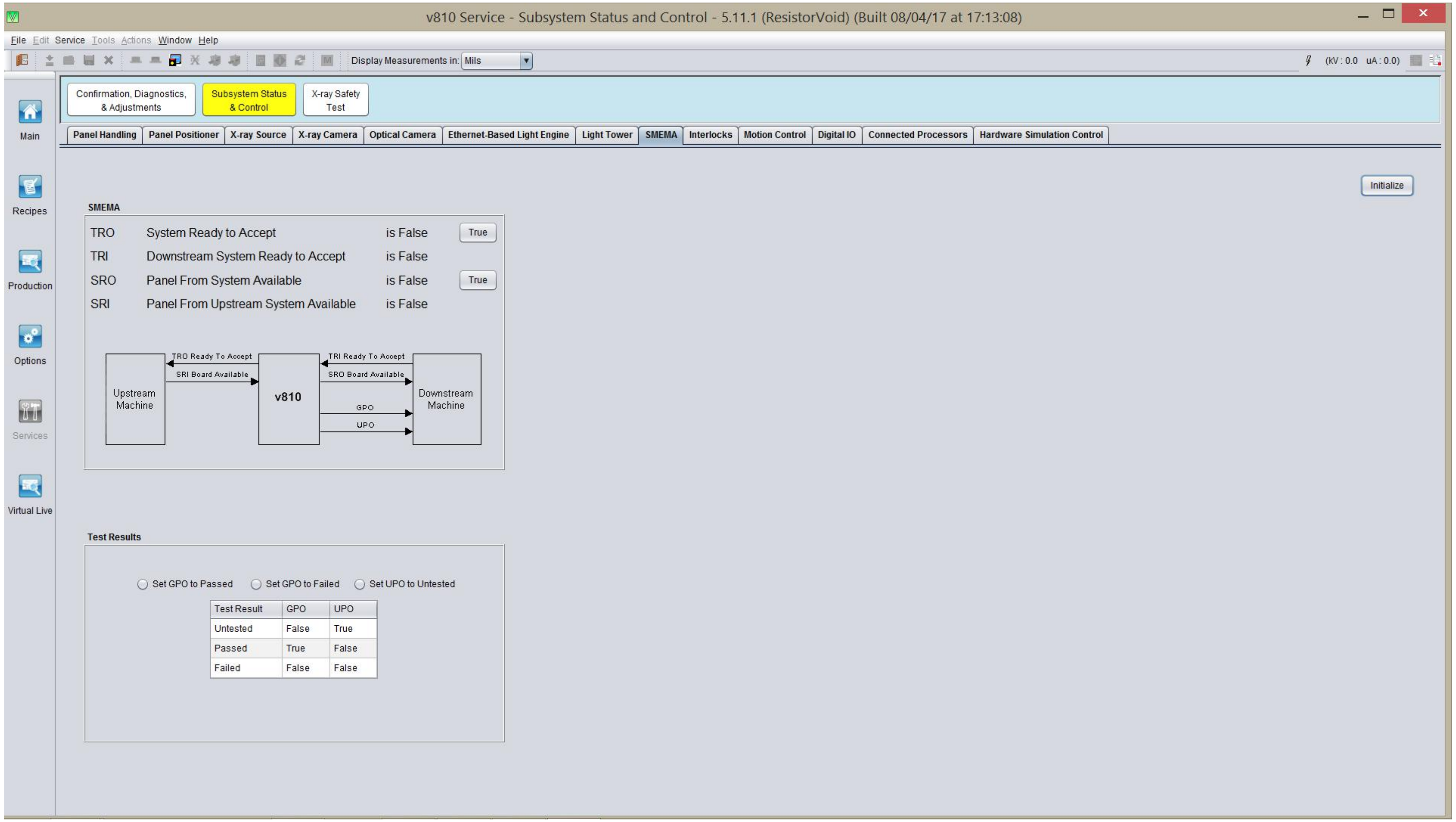
System GUI - Service tab - Subsystem Status & Control Page - SMEMA tab



➢ SMEMA connection verification

System GUI - Service tab - Subsystem Status & Control Page - Interlocks tab

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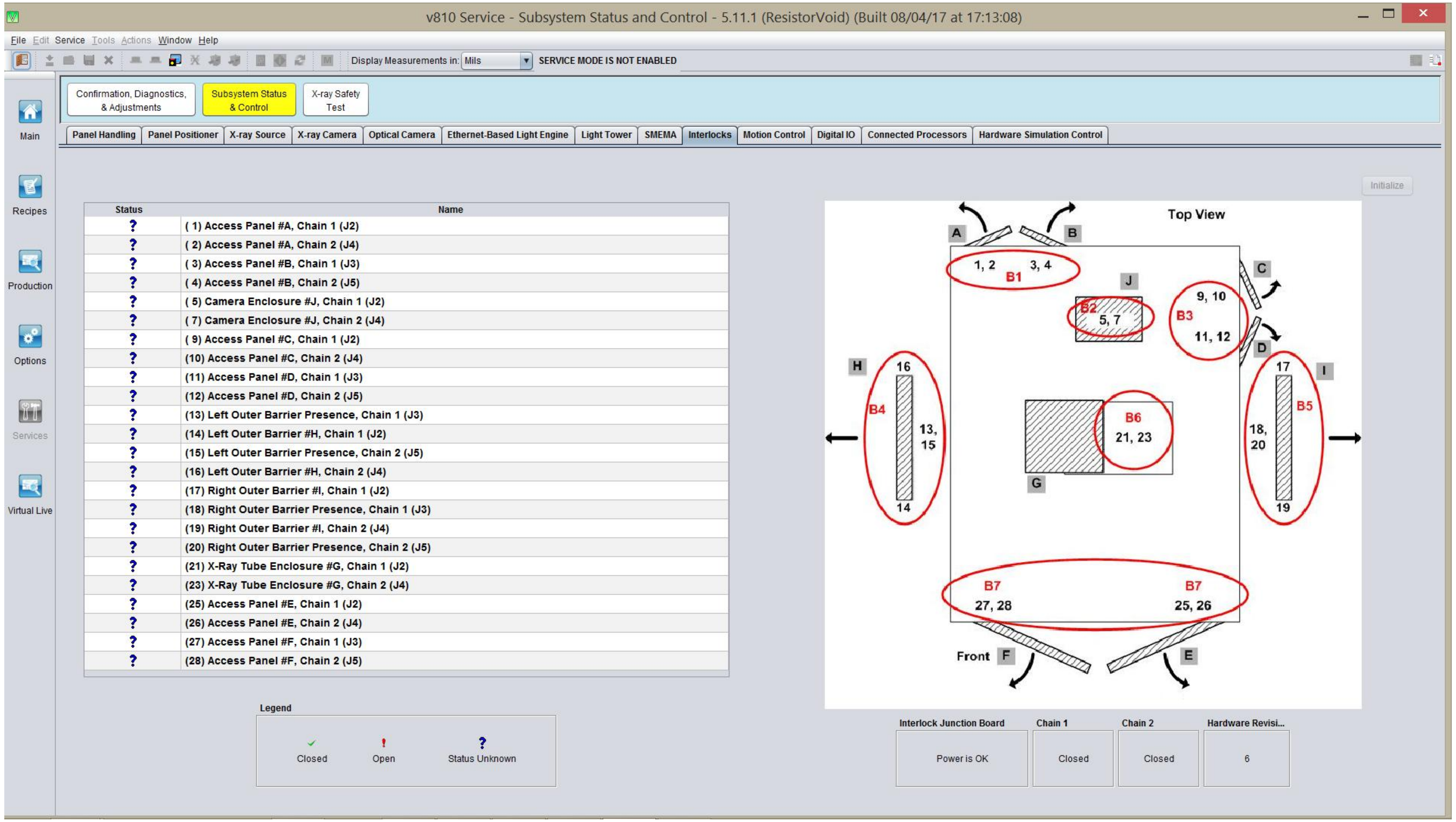
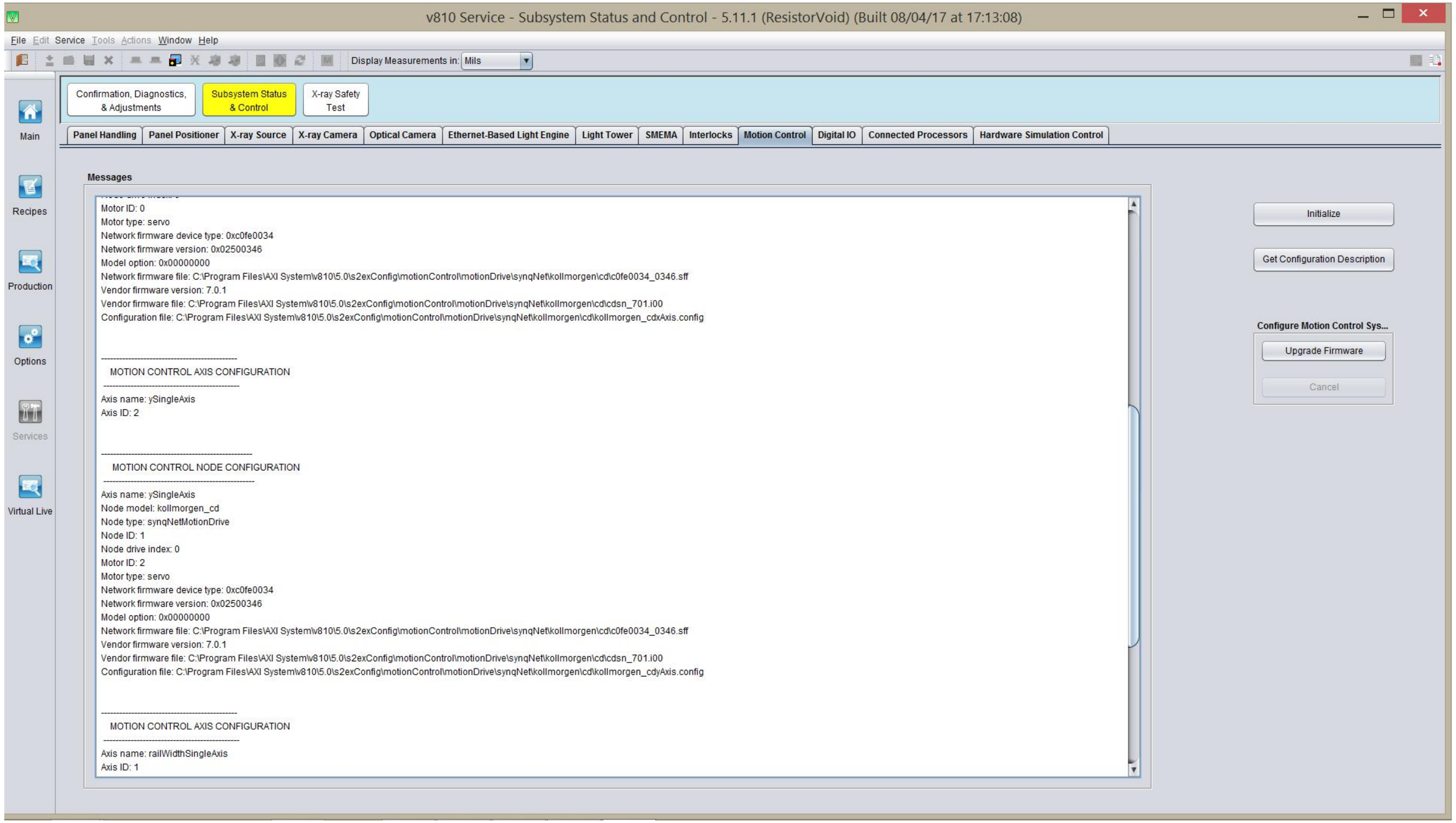
Automated Board Inspection



- System Interlock checking interface, when enable service mode, will trigger interlock break affected area
- Unable to initialize once DIO tab initialized. Direct switch to Service Mode will shown.

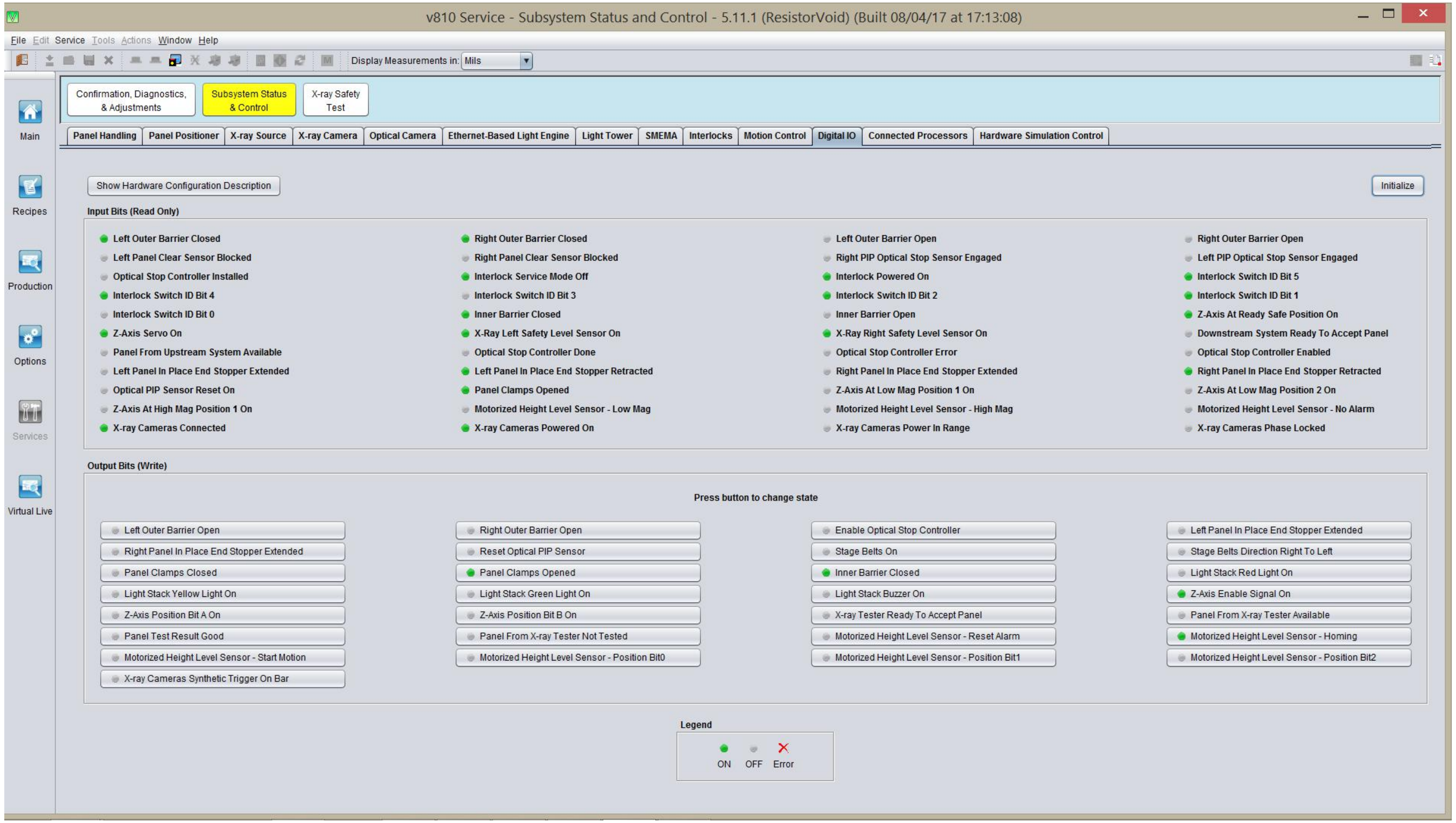
System GUI - Service tab - Subsystem Status & Control Page - Motion Control tab



- ➢ Check the connection for X and Y axis Kollmorgen kits with main server
- ➢ Perform Kollmorgen kits firmware upgrade (download motion profile to kollmorgen) each time change Kollmorgen kit or servo motor

ABI
Automated Board Inspection

System GUI - Service tab - Subsystem Status & Control Page - Digital I/O tab



- Direct control signal for acting cylinder and sensor input reading without checking interlock signal
- Unable initialize if Atom PC or SSCA didn’t initialize or do not power ON the device

System GUI - Service tab - Subsystem Status & Control Page - Connected Processors tab

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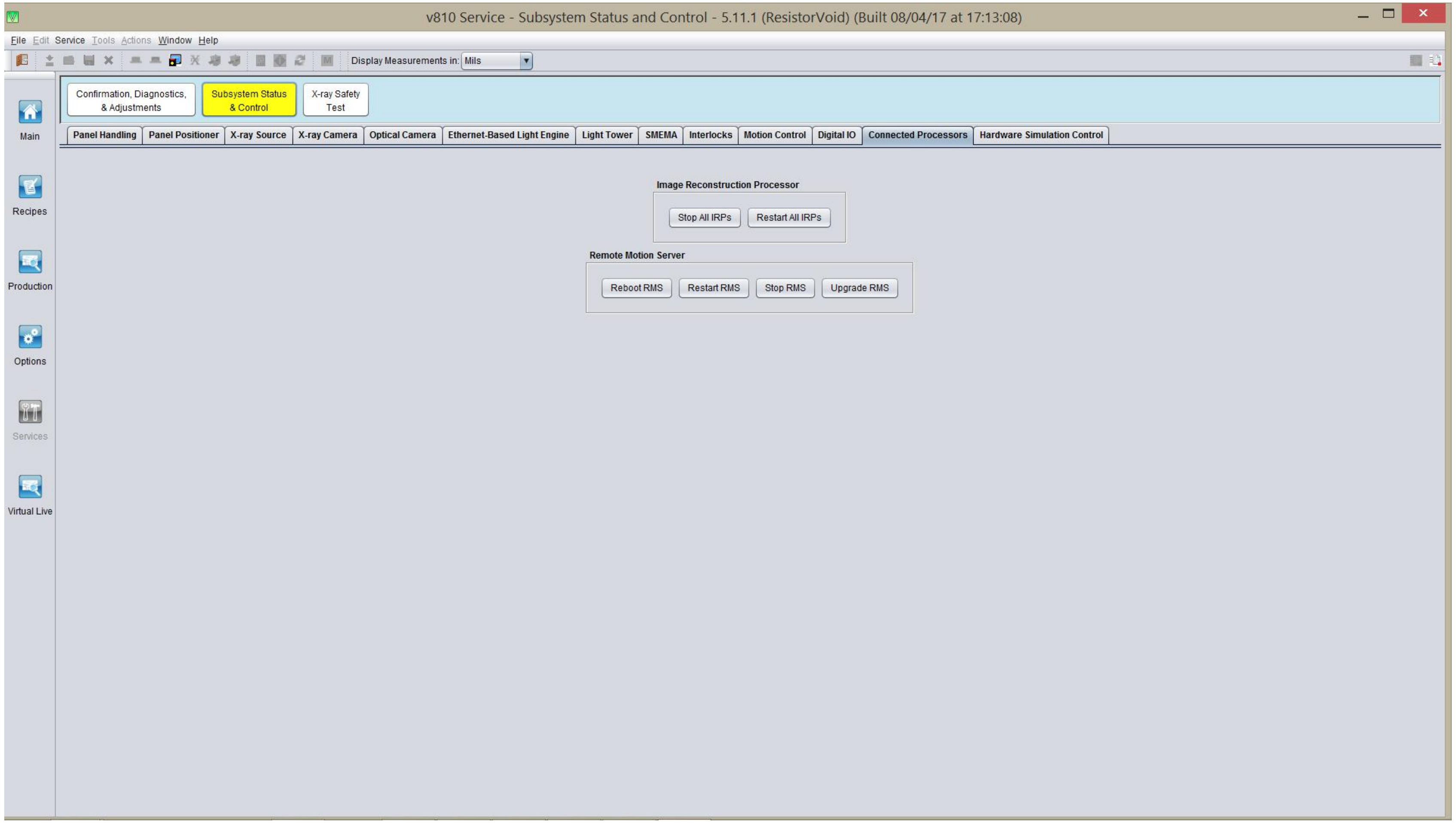
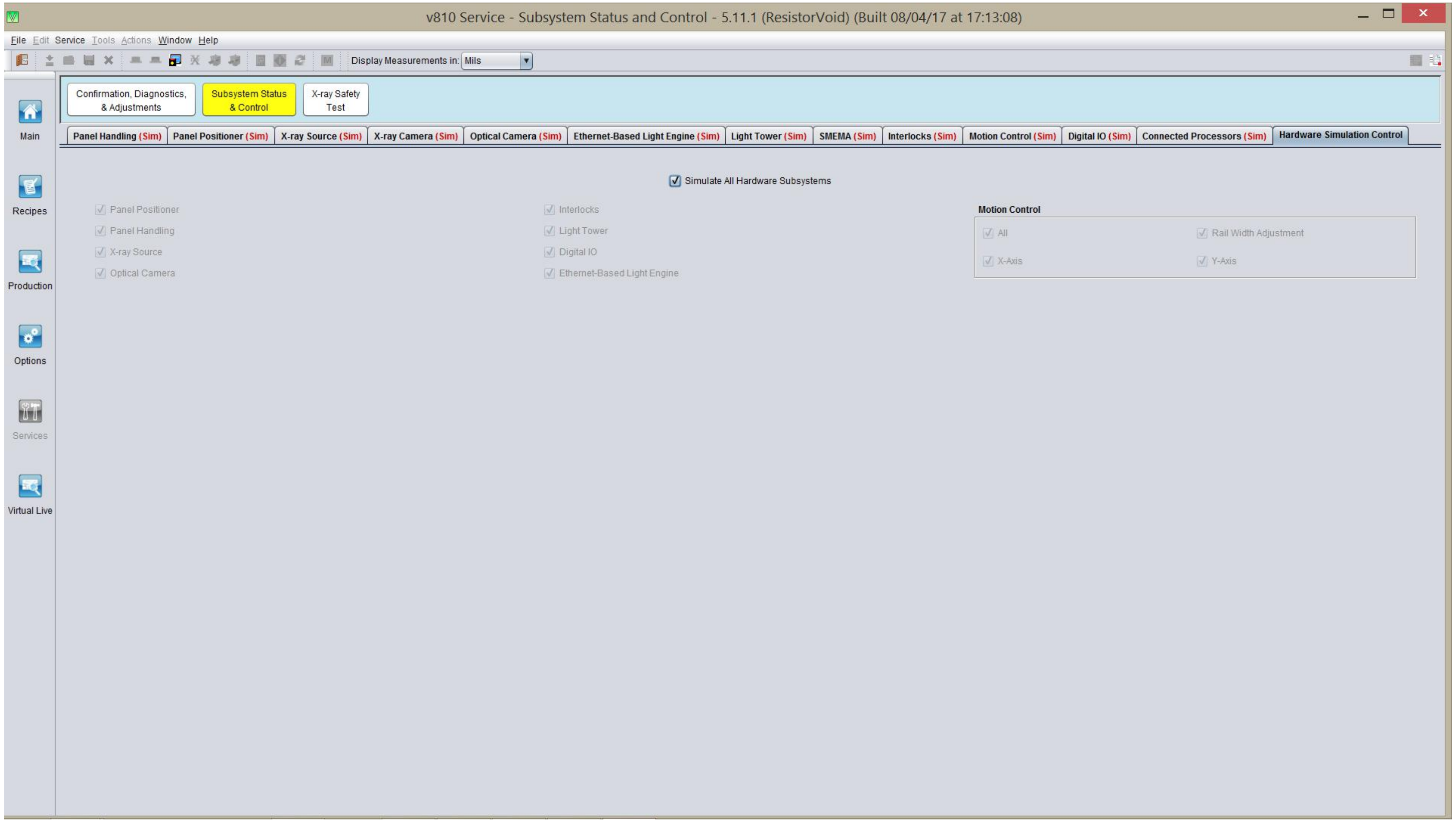


- Main controller tab to Stop/Restart and update RMS and IRP software
- Check communication between IRP, RMS(Atom PC) and main server connectivity
- Unable restart id devices didn't power On.

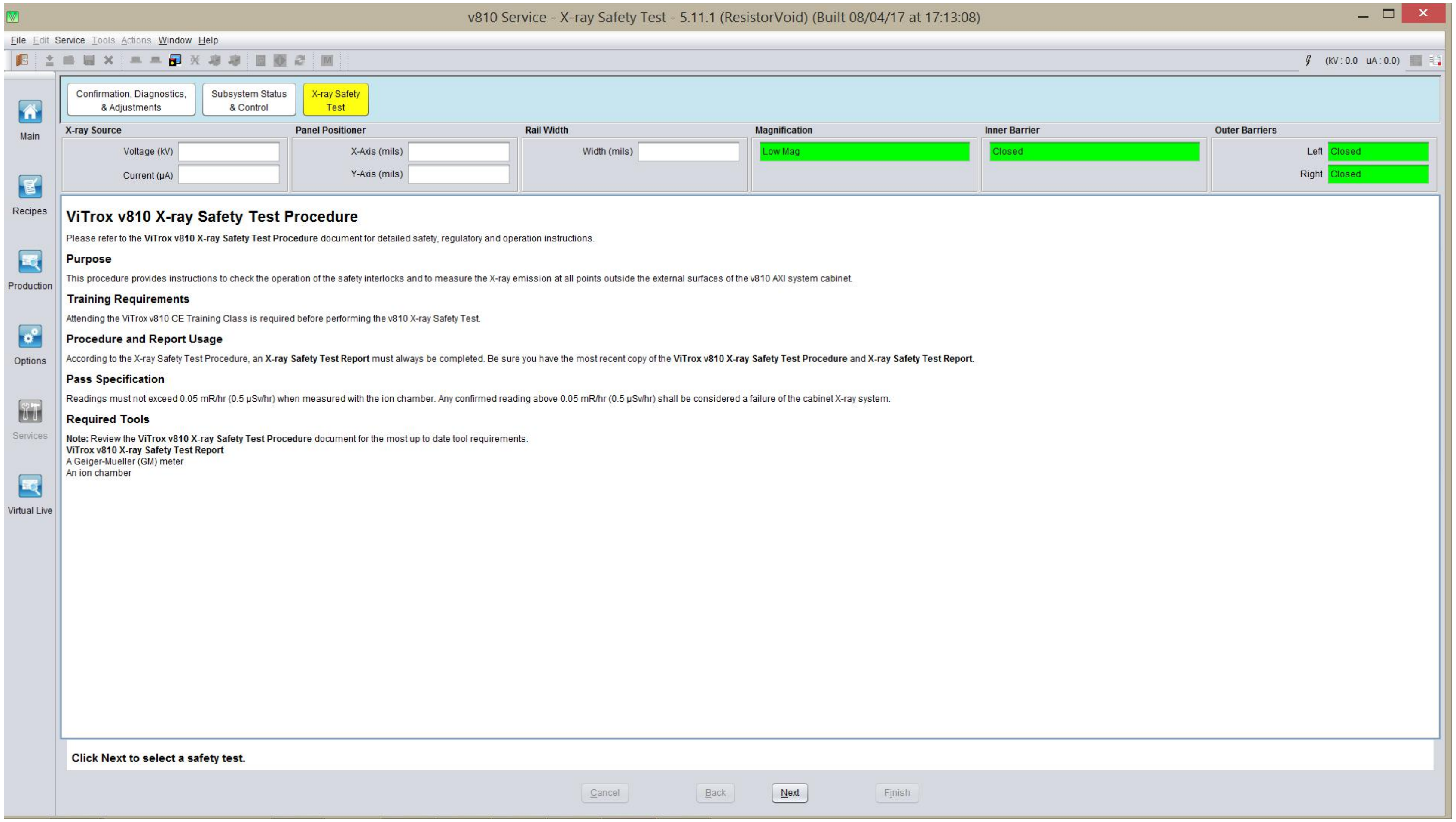
System GUI - Service tab - Subsystem Status & Control Page - Hardware Simulation tab



- Simulation hardware condition at Offline Learning Station (OLP)

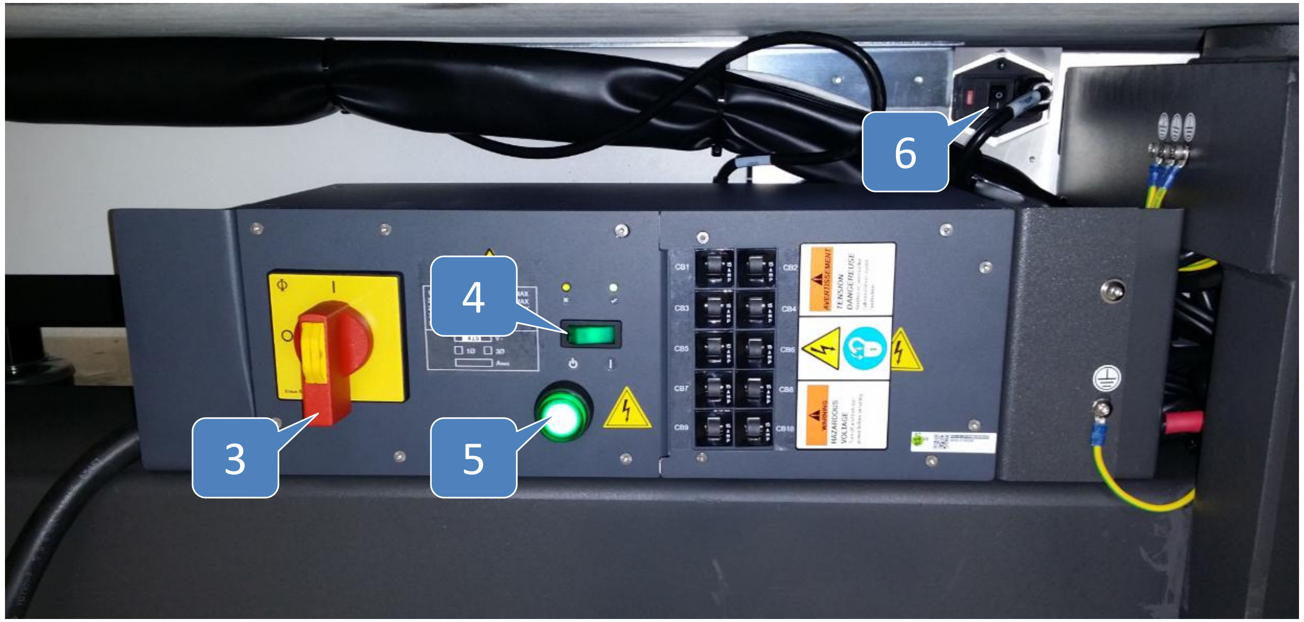
Automated Board Inspection

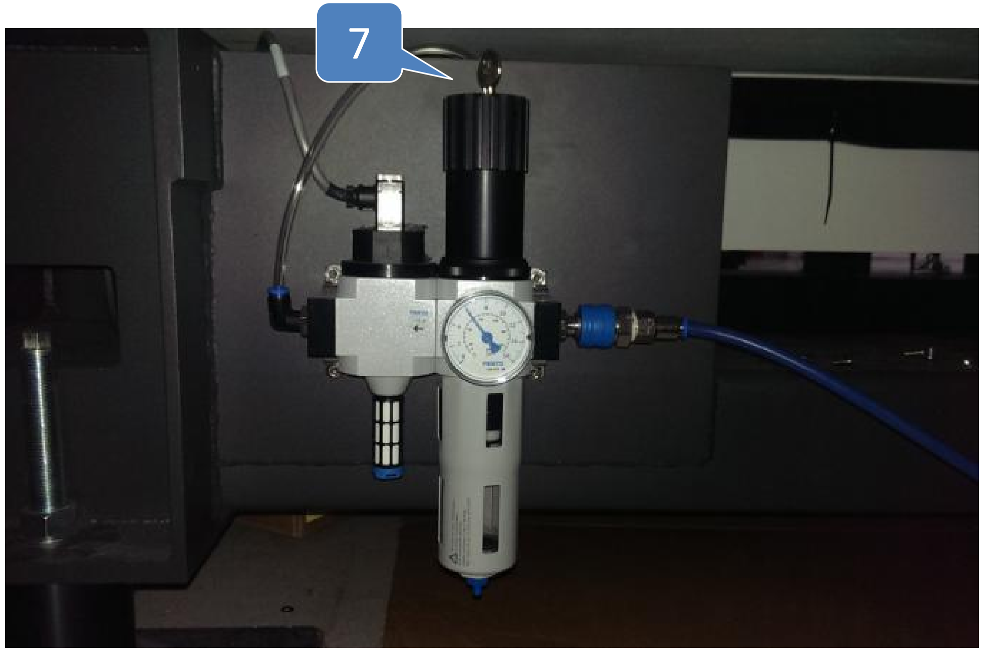
System GUI - Service tab - Xray Safety Test tab



- Main tab to perform X-Ray Survey with manual select test method, Low Power test and/or High Power Test

System Overview - System Start-up - PDU Power and Air Regulator Enable

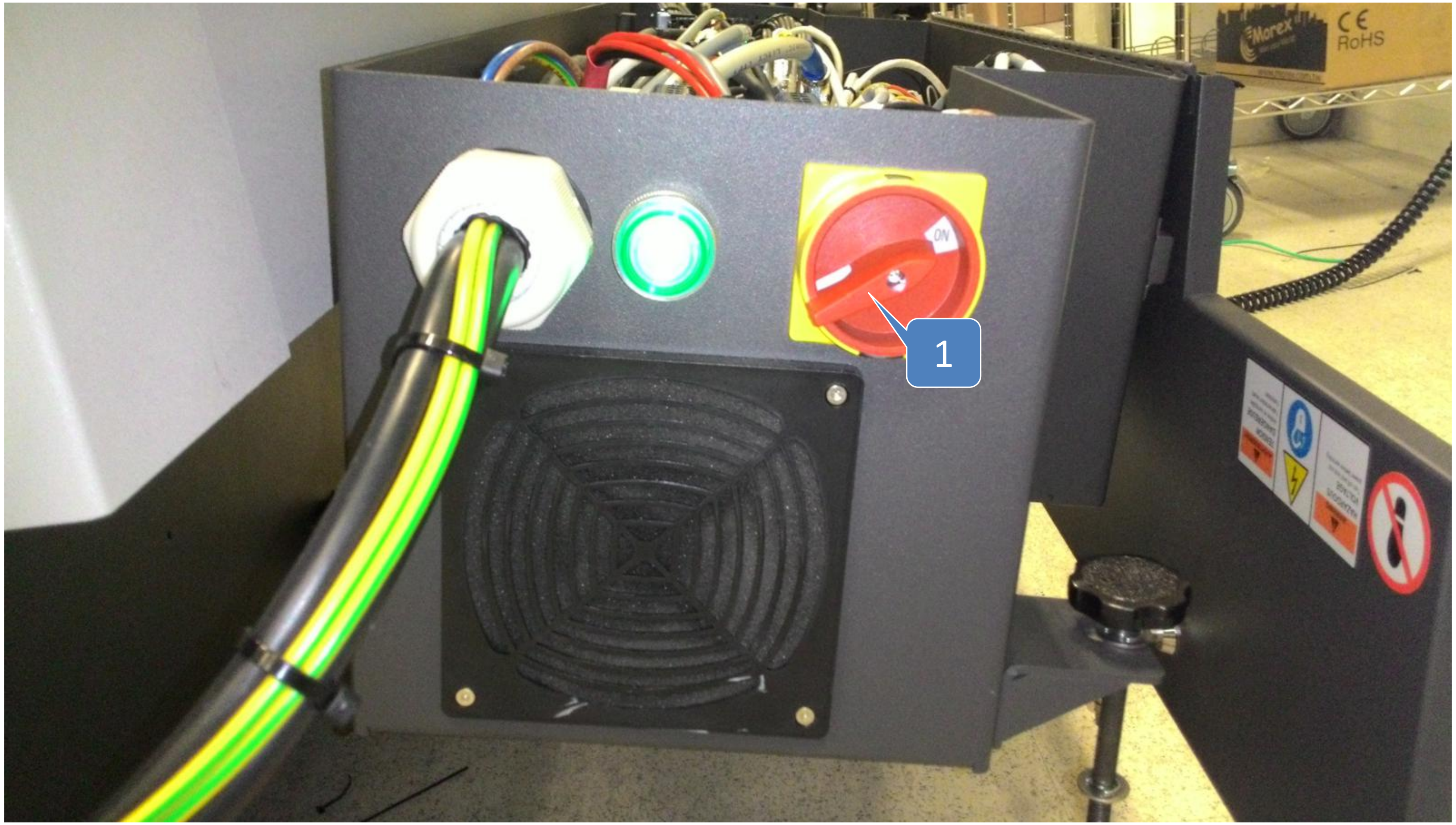




1. Release EMO (if enable) and ensure all access door closed
2. Power on incoming power at plant facilities
3. Switch main power inlet switch to “On” status
4. Turn On Unswitch port power on switch
5. Press “Power On” Button
6. Switch On Camera Controller Assemble Power
7. Set Air Inlet to 80 PSI (Lock key to turn knob and done, release lock)

System Overview - System Start-up - SSCA/MCA Box Power Enable

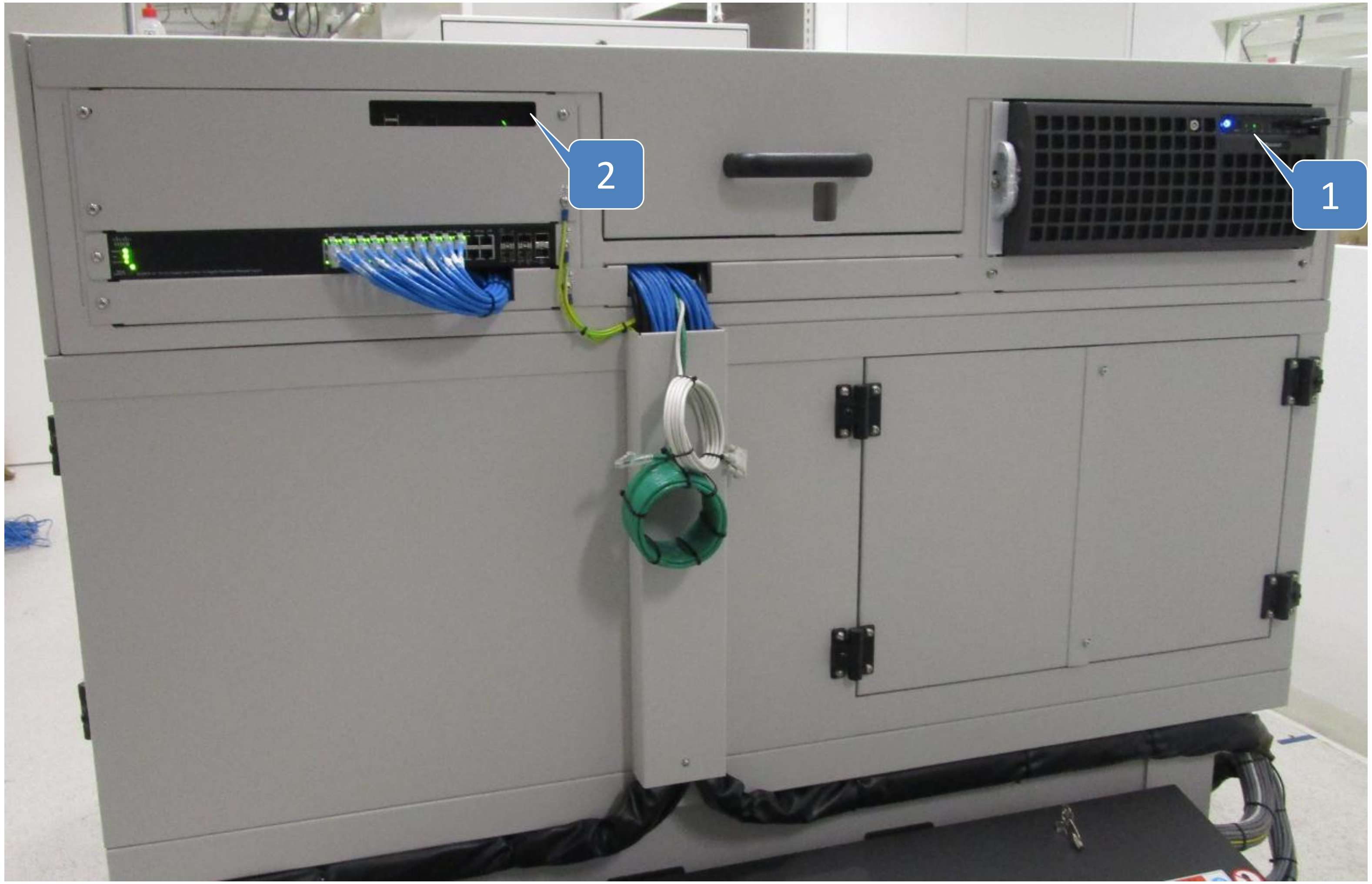
Automated Board Inspection



1. Switch SSCA/MCA Box main power inlet to “On” position

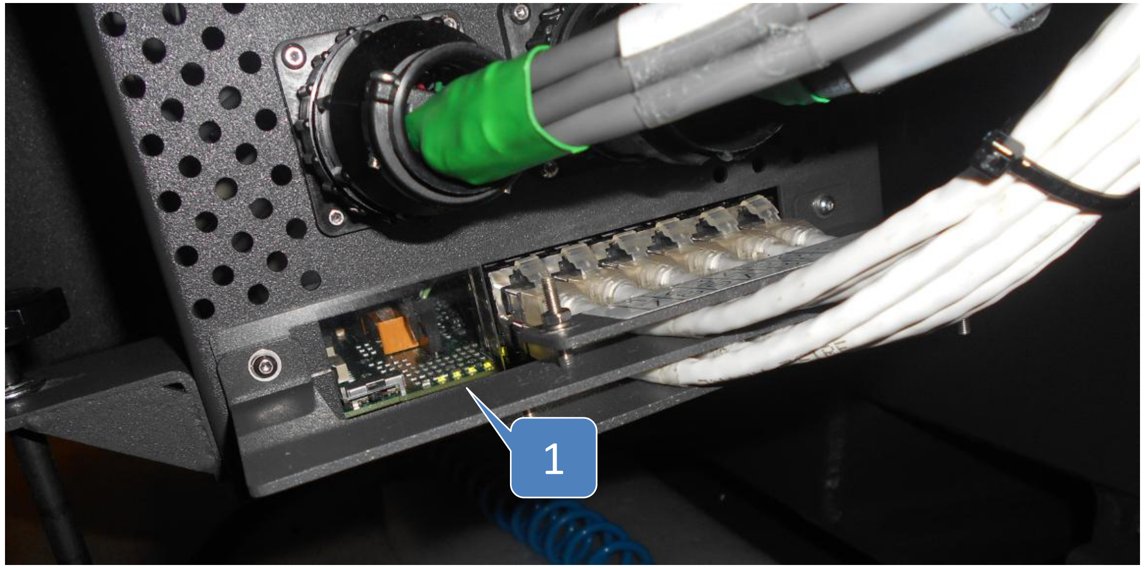
System Overview - System Start-up - Controllers(PC) Power On

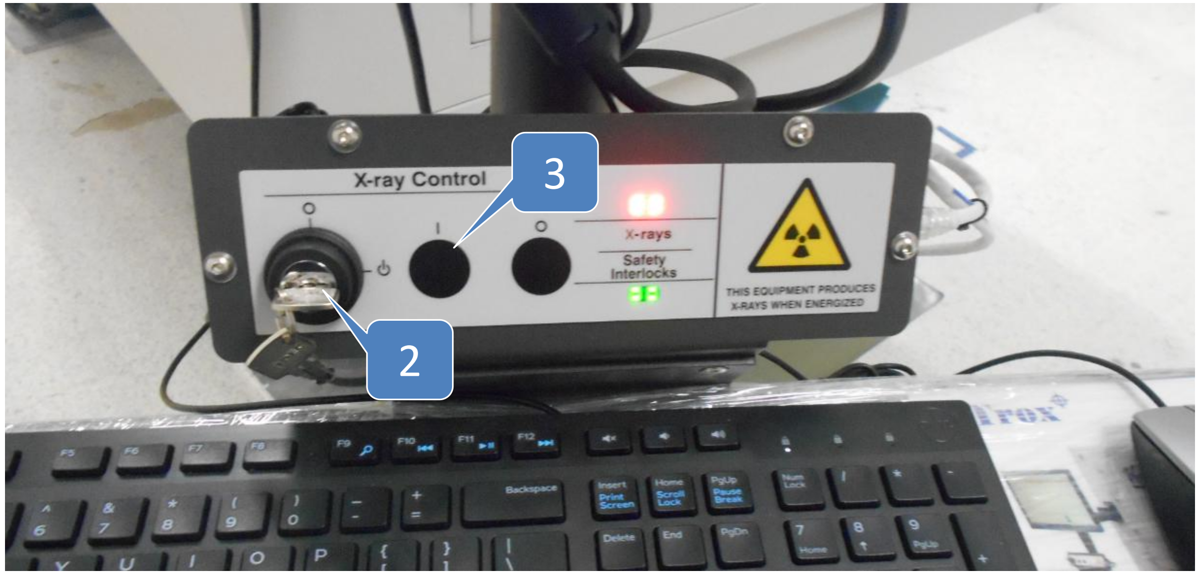
ABI
Automated Board Inspection



1. Turn on main controller
2. Turn on RMS (Atom PC)

System Overview - System Start-up - Operator Console Panel Activate





1. Visual check at Interlock card, 4 green LED should emitter once machine power on if no any interlock break.
2. Twish the key to “On” position.
3. Trigger the “X-Ray On” button and Red LED will emitter

System Overview - System Start-up - GUI Log in and system initialization

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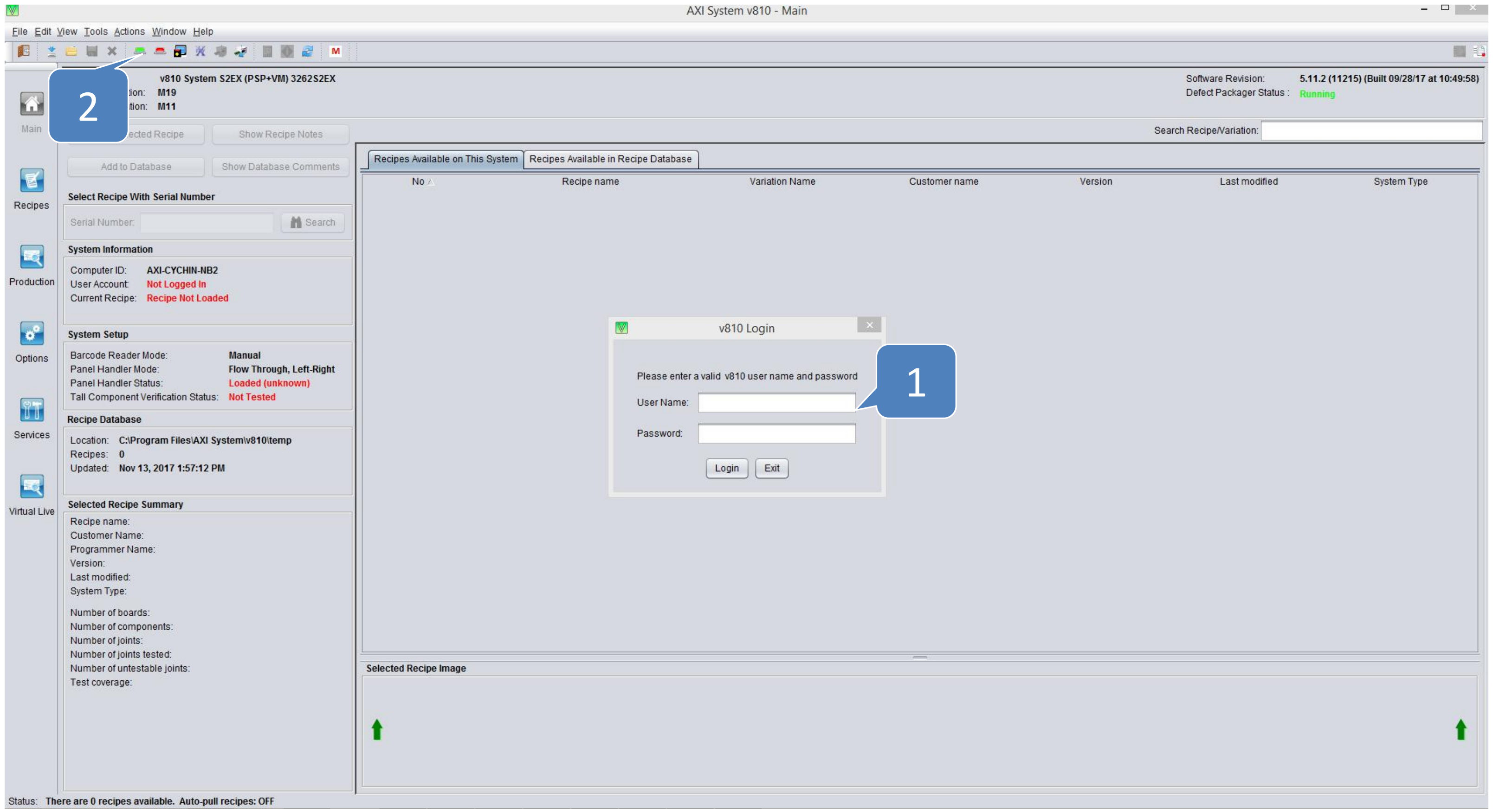

1. Launch V810 GUI and log in with desire user account and password
2. Click the start-up button or shortcut key “Ctrl+U” and wait for initialization complete

System Overview - System Start-up - X Ray Tube initialization Period

Step	Tube voltage (kV)	Tube current (µA)	Warm-up time (minutes)		
			Pattern 1	Pattern 2	Pattern 3
			8 hours to 1 month after power off	1 to 3 months after power off	3 months or more after power off
1	27	0	1	5	10
2	54	30	1	5	30
3	81	90	3	6	20
4	108	150	3	7	30
5	121	220	3	7	20
6	130	300	4	10	10
Total (minutes)			15	40	120

- During machine start-up X-ray tube will self perform warm-up operation
- Warm-up operation pattern varies according to the period X-ray tube had been off
- X-ray tube do have controller board itself and shielded
- Warm-up operation will self define by the X-ray tube

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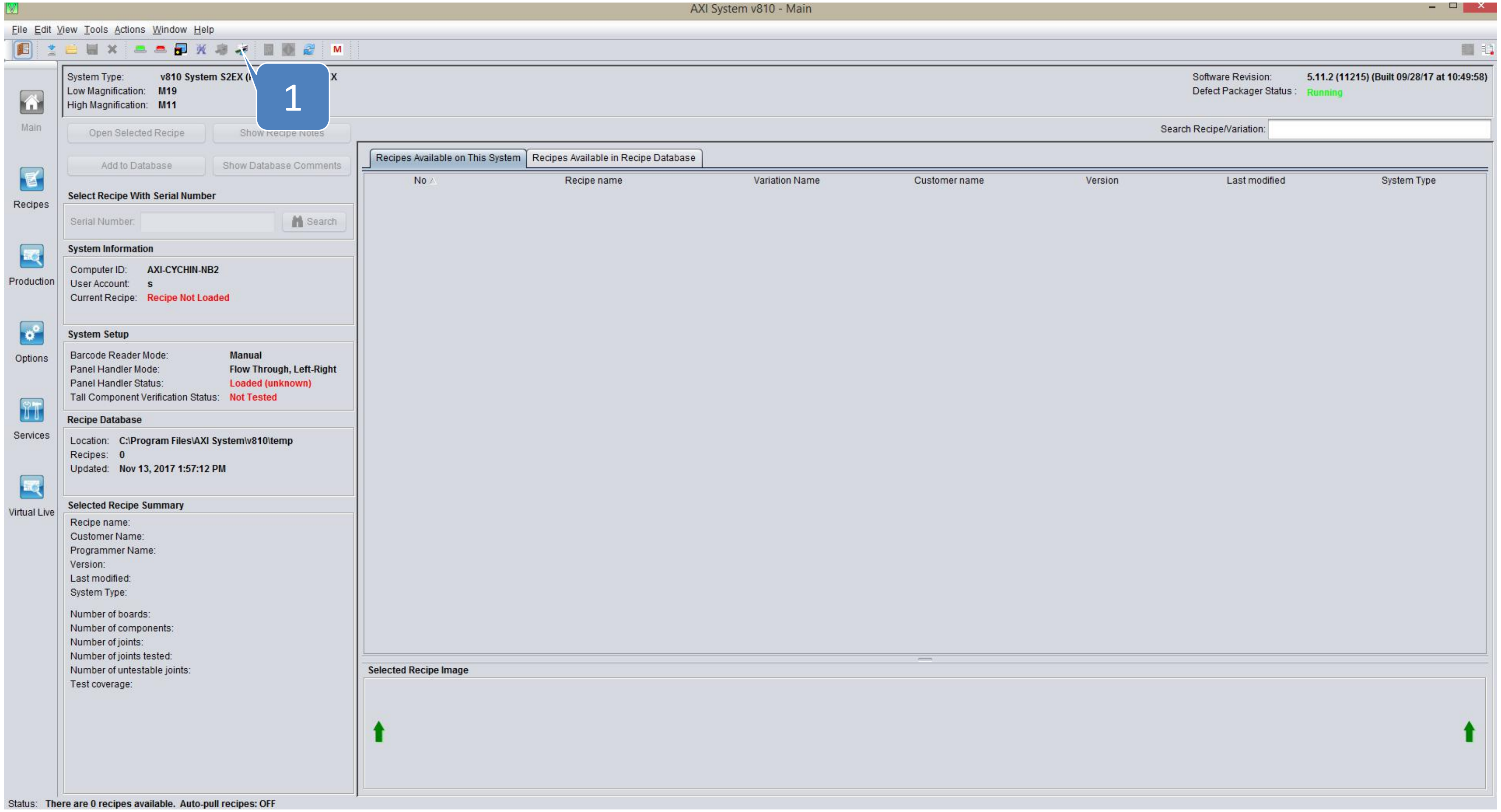
System Overview - System Shut Down - Unload Panel



1. At main page , click Unload Icon to unload panel from system before machine shutdown. (this is importance to avoid board crush or system error during next start-up)

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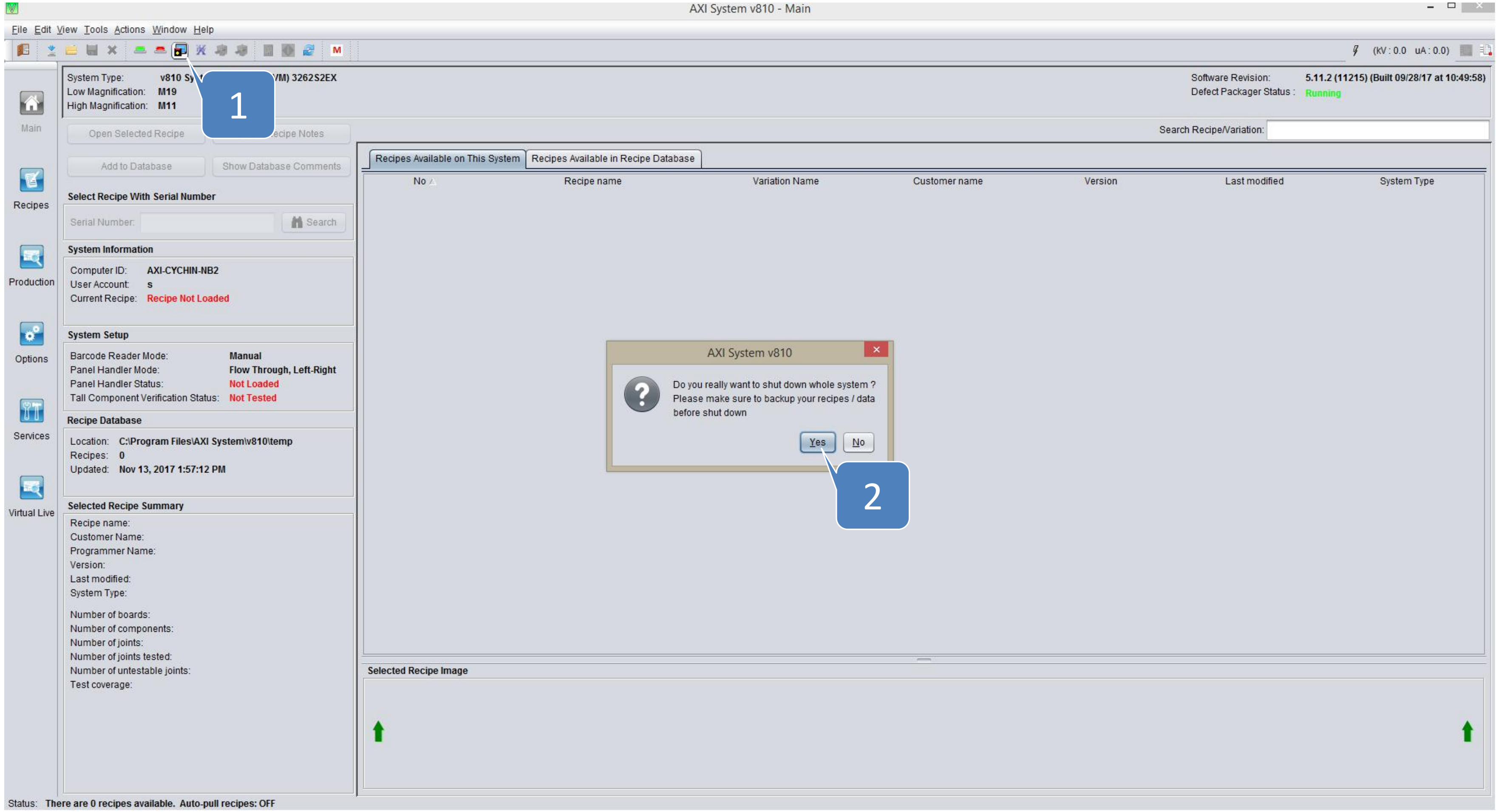
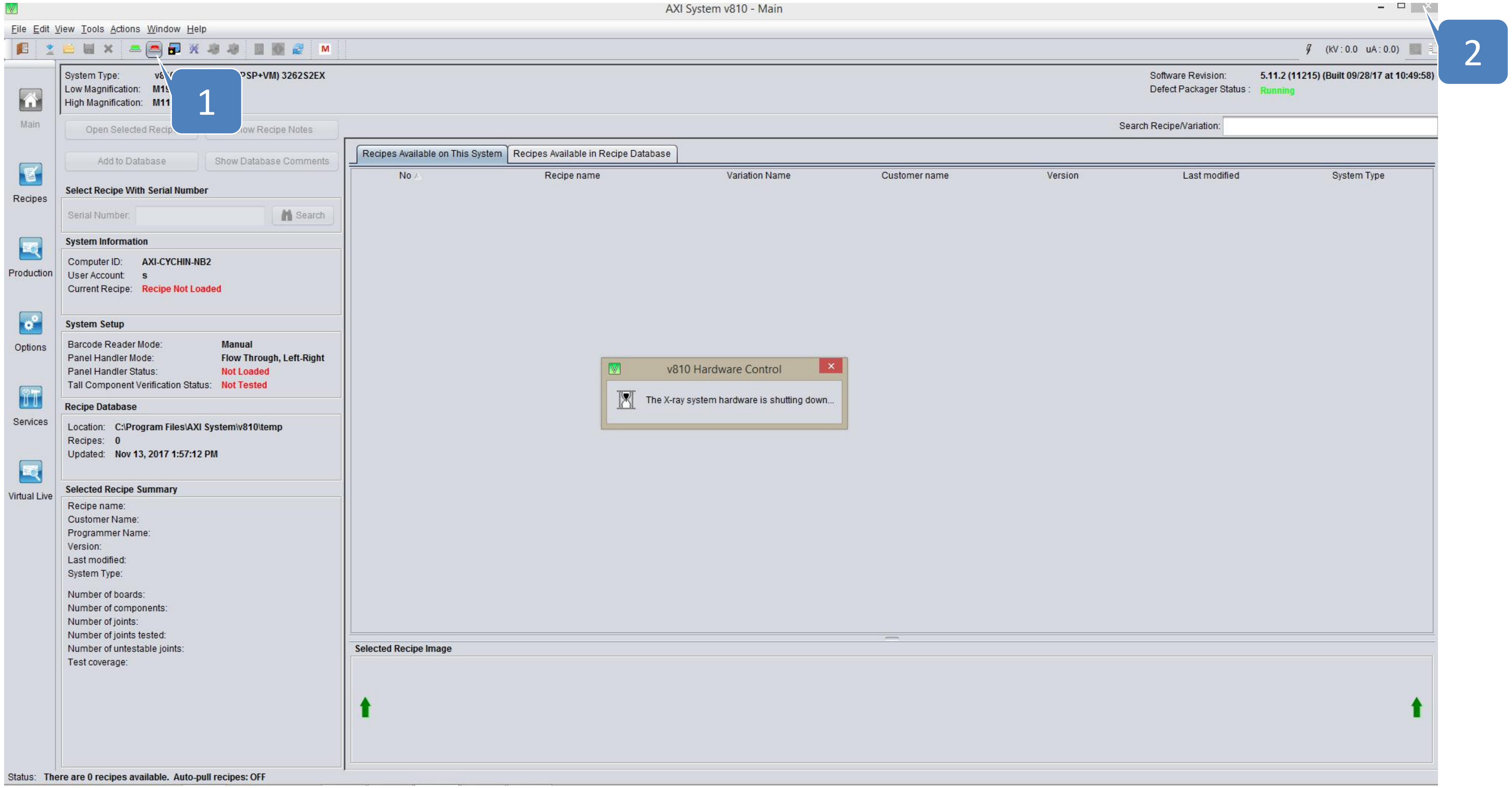
System Overview - System Shut Down - Whole System Shutdown



1. Click “Shut Down whole system” icon to completely power off machine including Atom PC.
   Remark, this **feature only available for software version 5.11.2 and above.**
2. Click “Yes” to continues and system will force shutdown main controller and RMS (Atom PC) without save other running task in progress.
3. Once done, may proceed turn off Xray source at Operator Console, main power input, SSCA power, press EMO and turn off air pressure inlet

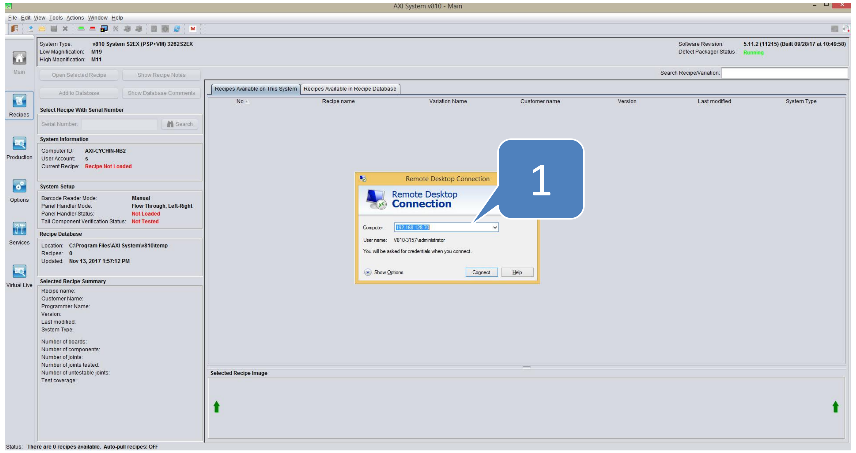
System Overview - System Shut Down - System Shutdown - GUI

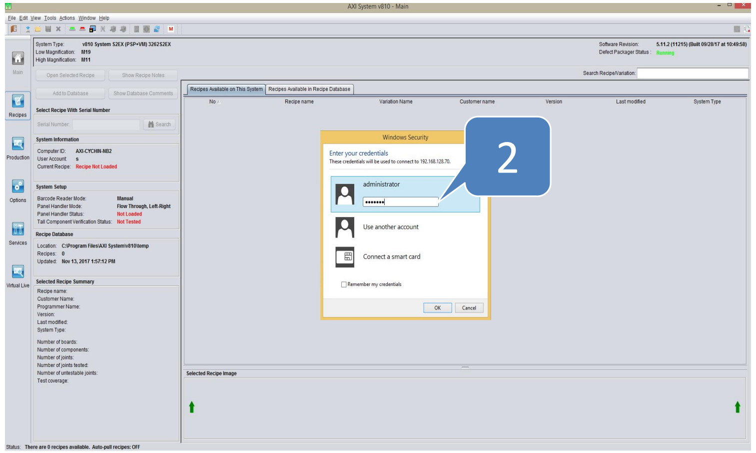


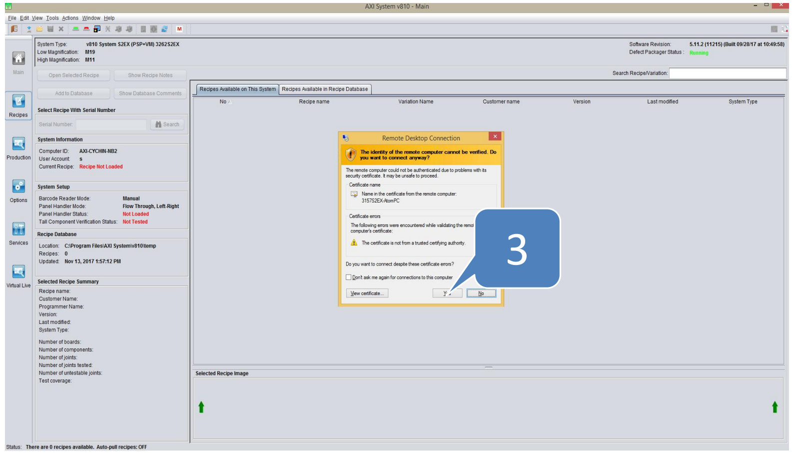
1. Click “Shut Down” icon to turn off X-ray sources and deactivate system hardware peripheral.
2. Exit the V810 GUI

System Overview - System Shut Down - System Shutdown - RMS (Atom PC)

Automated Board Inspection

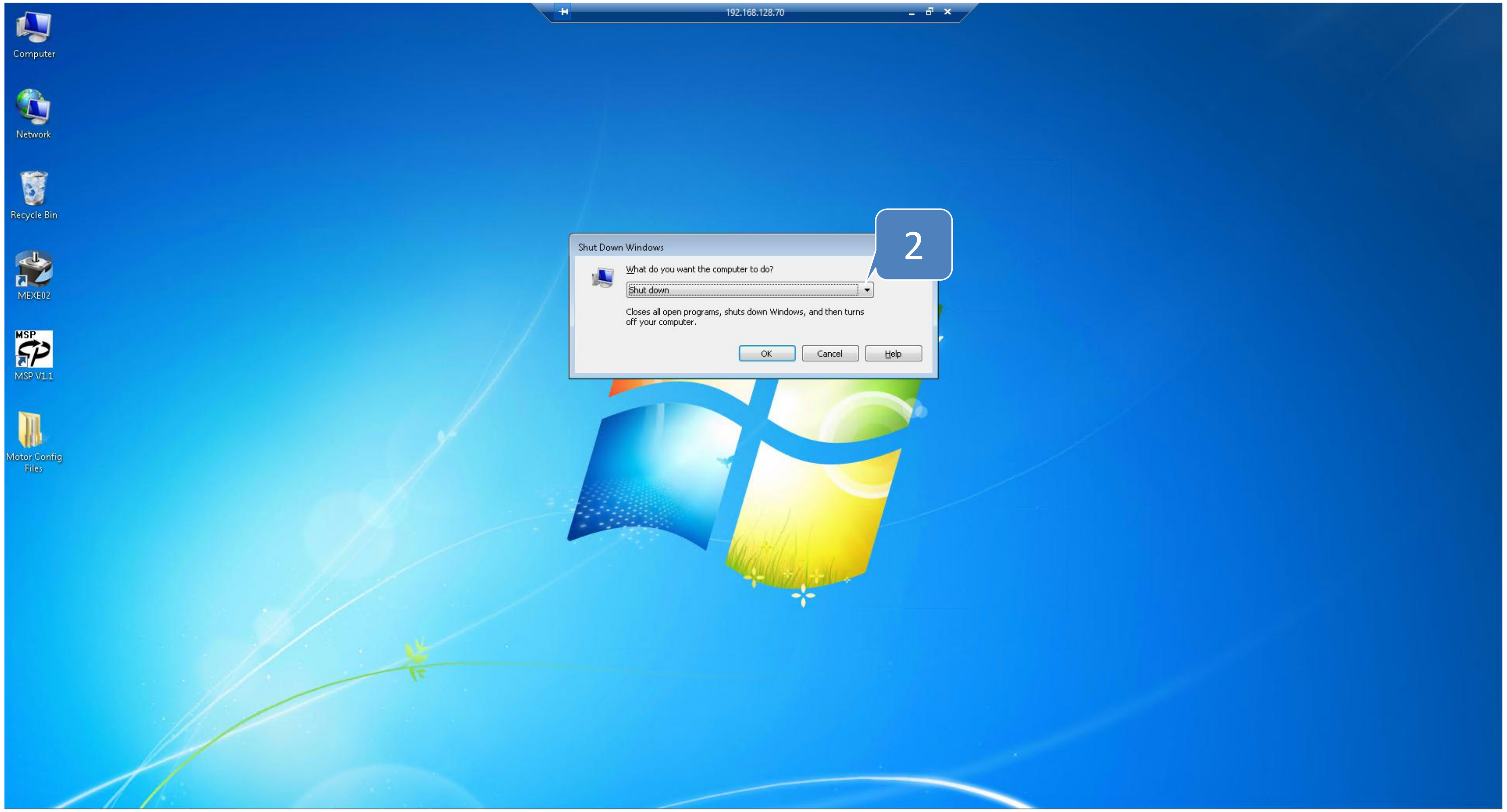






1. Launch “Remote Desktop Connection” from main controller and remote to Atom PC by using IP “192.168.128.70”
2. Log in with user “Administrator” & Password “Please!”
3. Click “Yes” to proceed

System Overview - System Shut Down - System Shutdown - RMS(Atom PC)



1. Press “Ctrl+F4” and action tab will shown
2. Select “Shutdown” and press “OK”, Atom PC will be automatically shutdown after few second.
3. Back to main controller, shut down PC as usual
4. Once done, may proceed turn off Xray source at Operator Console, main power input, SSCA power, press EMO and turn off air pressure inlet

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